

Prob_ode

January 31, 2022

```
[1]: # Make inline plots vector graphics instead of raster graphics
%matplotlib inline
from IPython.display import set_matplotlib_formats

set_matplotlib_formats("pdf", "svg")

# Plotting
import matplotlib.pyplot as plt
```

```
[2]: import numpy as np
from probnum.diffeq import probsolve_ivp
```

1 2-D damped oscillator (Linear)

```
[3]: def f(t, y):
    y1, y2 = y
    return np.array([-0.1*y1+2*y2, -2*y1-0.1*y2])

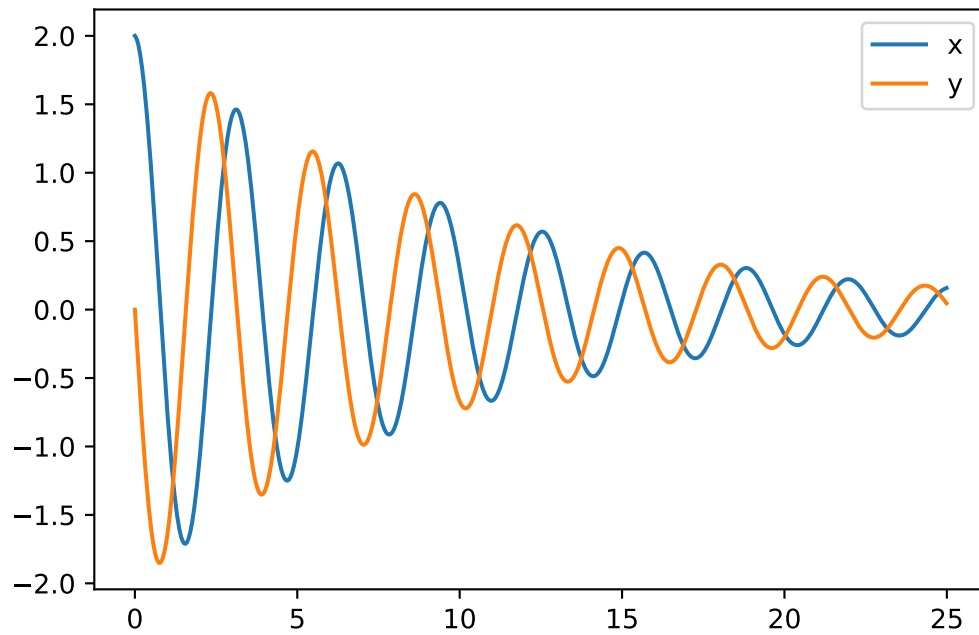
def df(t, y):
    y1, y2 = y
    return np.array([[ -0.1, 2], [-2, -0.1]])

t0 = 0.0
tmax = 25.0
y0 = np.array([2, 0])
```

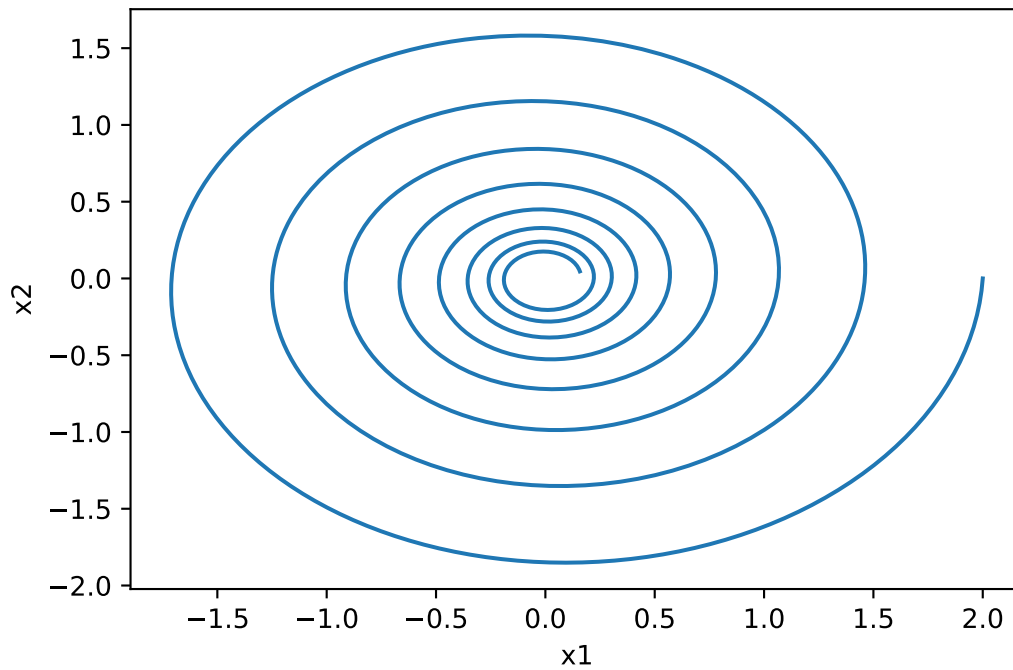
```
[4]: # IBM(1) with EK0, solving an IVP with adaptive steps works.
sol = probsolve_ivp(f, t0, tmax, y0, df=df, algo_order=4, method="EK1")
dt=0.01
evalgrid = np.arange(0, 25, dt)
means = sol(evalgrid).mean

plt.plot(evalgrid, means[:, 0], label="x")
```

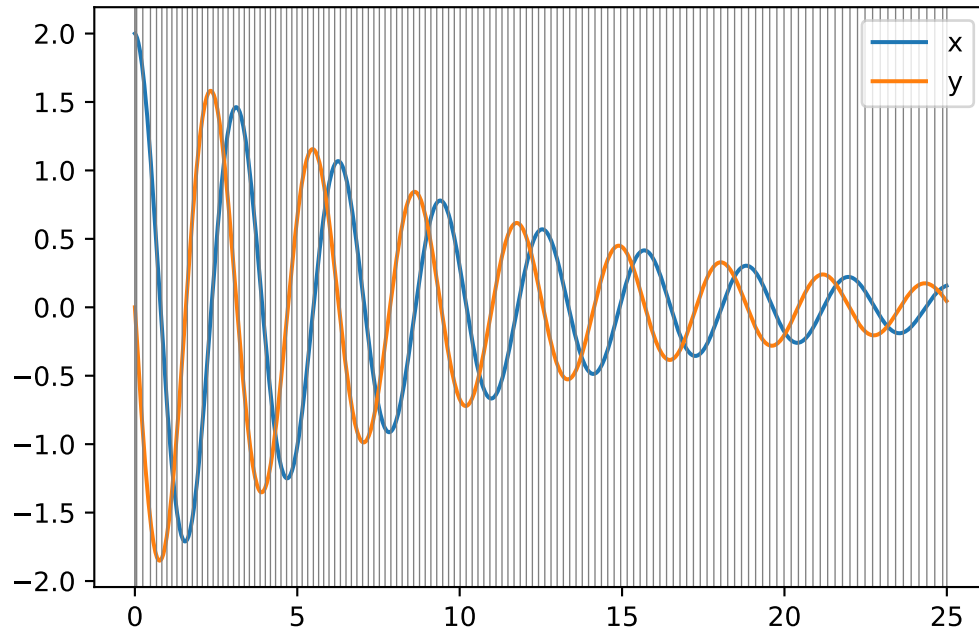
```
plt.plot(evalgrid, means[:, 1], label="y")
plt.legend(loc="upper right")
plt.show()
```



```
[5]: plt.plot(means[:, 0], means[:, 1])
plt.xlabel("x1")
plt.ylabel("x2")
plt.show()
```



```
[6]: # Let's visualise the individual steps.
plt.plot(evalgrid, means[:, 0], label="x")
plt.plot(evalgrid, means[:, 1], label="y")
for t in sol.locations:
    plt.axvline(t, linewidth=0.5, color="gray")
plt.legend(loc="upper right")
plt.show()
```



```
[7]: algo_orders = [2, 3]
filters = ["EK0", "EK1"]

fig, ax = plt.subplots(
    nrows=len(algo_orders),
    ncols=len(filters),
    sharex=True,
    sharey=True,
    figsize=(12, 12),
)

evalgrid = np.linspace(t0, tmax, 100)

for idx in range(len(algo_orders)): # algo_orders in rows
    for jdx in range(len(filters)): # filters in cols
        sol = probsolve_ivp(
            f,
            t0,
            tmax,
            y0,
            df=df,
            algo_order=algo_orders[idx],
            method=filters[jdx],
        )
```

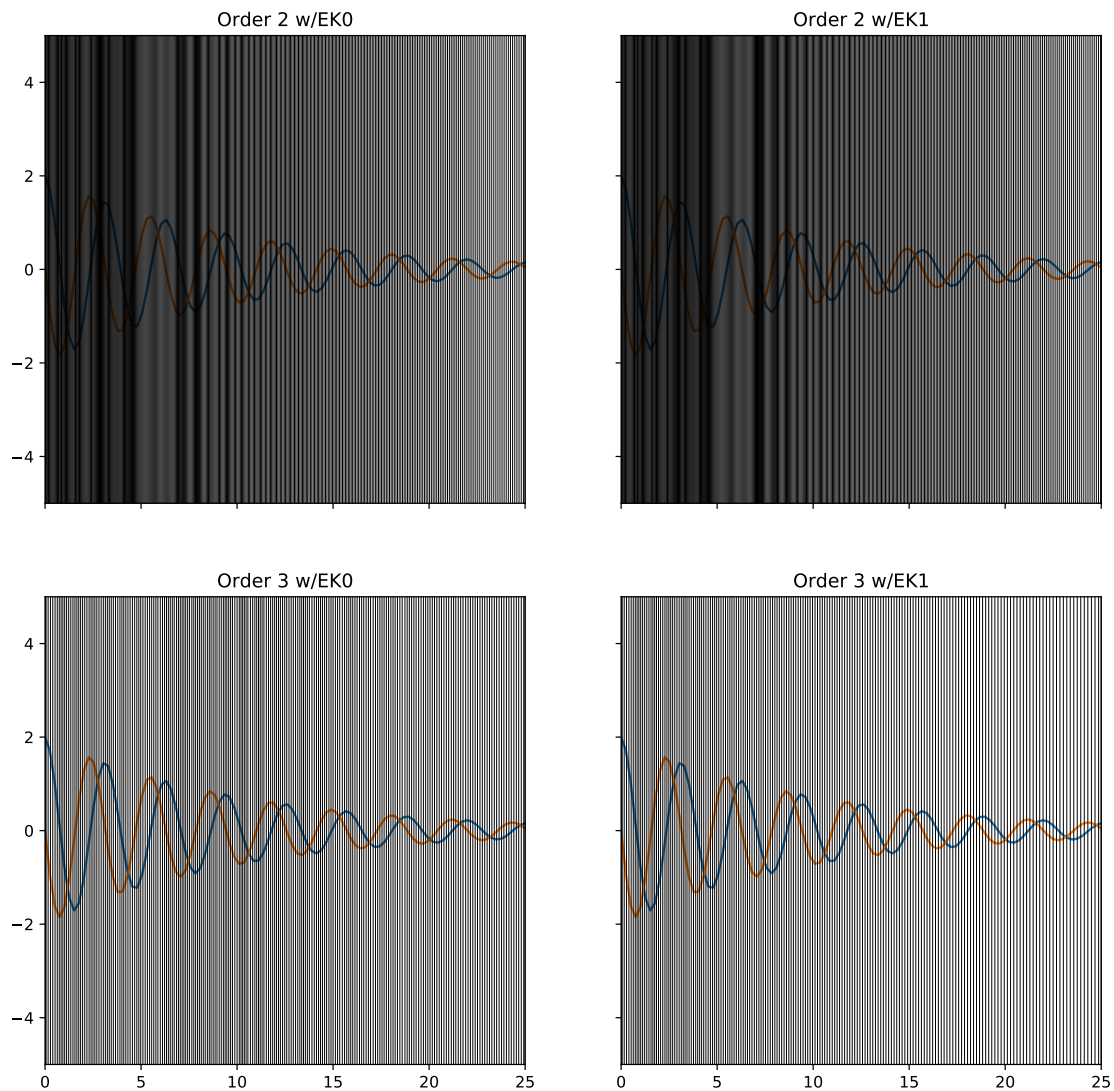
```

solution = sol(evalgrid)
ts, means, stds = evalgrid, solution.mean, solution.std
ax[idx][jdx].plot(ts, means)

for t in sol.locations:
    ax[idx][jdx].axvline(t, linewidth=0.2, color="black")
ax[idx][jdx].set_title(f"Order {algo_orders[idx]} w/{filters[jdx]}")
ax[idx][jdx].set_xlim((t0, tmax))
ax[idx][jdx].set_ylim((-5, 5))

plt.show()

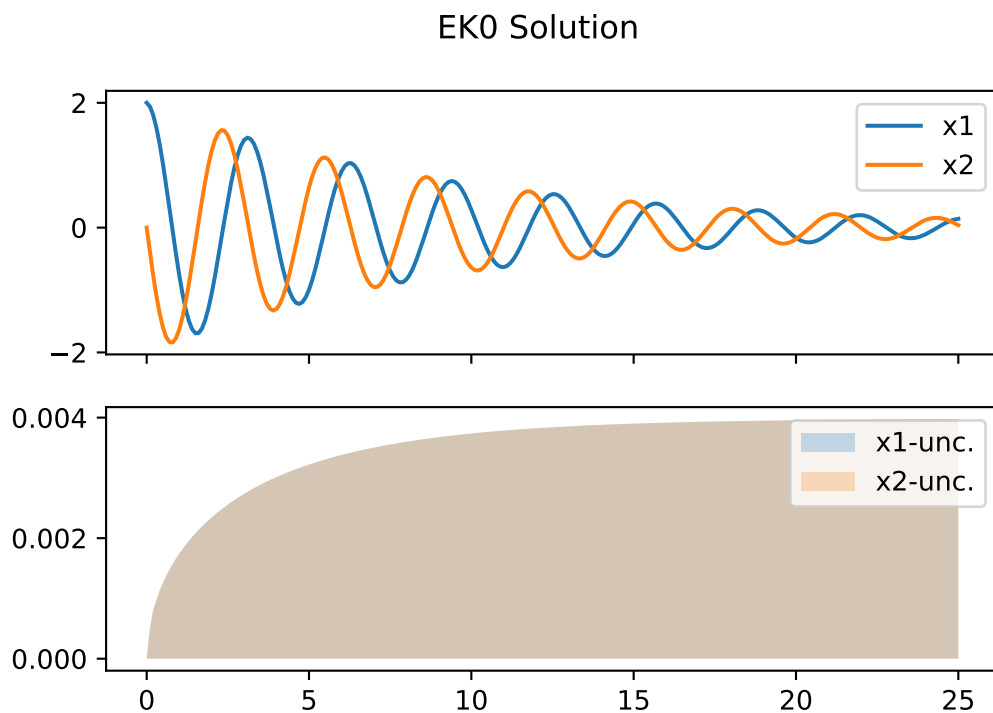
```



```
[8]: sol = probsolve_ivp(
    f, t0, tmax, y0, df=df, step=0.1, adaptive=False, diffusion_model="dynamic"
)

means, stds = sol.states.mean, sol.states.std

fig, (ax1, ax2) = plt.subplots(nrows=2, ncols=1, sharex=True)
ax1.plot(sol.locations, means[:, 0], label="x1")
ax1.plot(sol.locations, means[:, 1], label="x2")
ax1.legend()
ax2.fill_between(sol.locations, stds[:, 0], alpha=0.25, label="x1-unc.")
ax2.fill_between(sol.locations, stds[:, 1], alpha=0.25, label="x2-unc.")
ax2.legend()
fig.suptitle("EK0 Solution")
plt.show()
```



```
[9]: sol = probsolve_ivp(
    f,
    t0,
    tmax,
    y0,
    df=df,
    step=0.5,
```

```

    algo_order=1,
    adaptive=False,
    diffusion_model="dynamic",
)

```

```

[10]: import numpy as np

from probnum.diffeq import probsolve_ivp

rng = np.random.default_rng(seed=123)

num_samples = 150
locations = np.arange(0.0, 22.1, 0.05)
locations = sol.locations
samples = sol.sample(rng=rng, t=locations, size=num_samples)

solution = sol(locations)
means = solution.mean
stds = solution.std

```

```

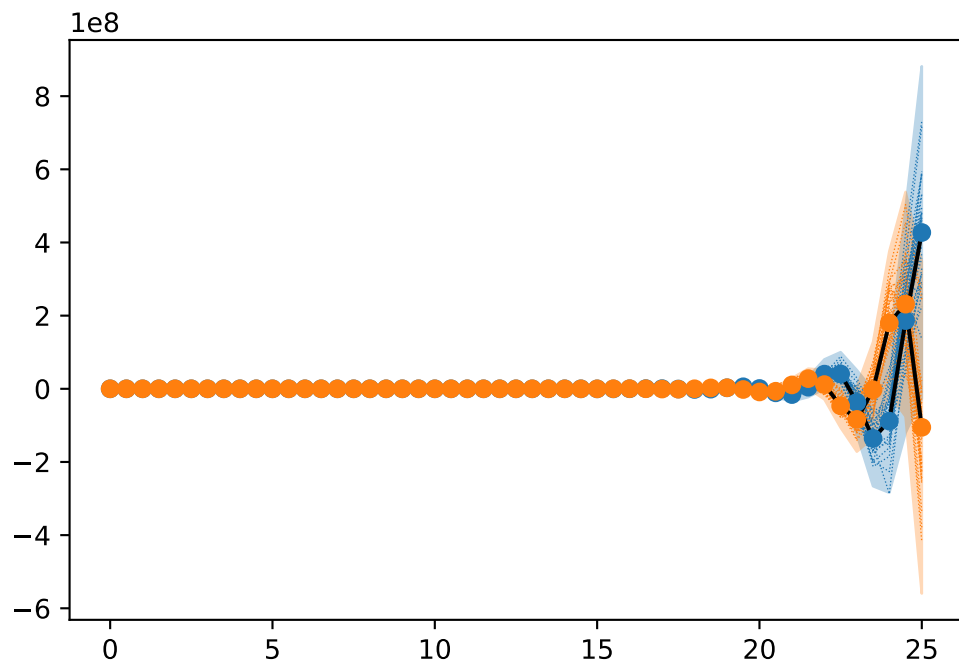
[11]: fig, (ax1) = plt.subplots(nrows=1, ncols=1, sharex=True)
for sample in samples[::5]:
    ax1.plot(locations, sample[:, 0], ":", color="C0", linewidth=0.5, alpha=1)
    ax1.plot(locations, sample[:, 1], ":", color="C1", linewidth=0.5, alpha=1)
ax1.plot(locations, means[:, 0], label="x1", color="k")
ax1.plot(locations, means[:, 1], label="x2", color="k")
ax1.fill_between(
    locations,
    means[:, 0] - 3 * stds[:, 0],
    means[:, 0] + 3 * stds[:, 0],
    color="C0",
    alpha=0.3,
)
ax1.fill_between(
    locations,
    means[:, 1] - 3 * stds[:, 1],
    means[:, 1] + 3 * stds[:, 1],
    color="C1",
    alpha=0.3,
)

ax1.plot(sol.locations, sol.states.mean, "o")
fig.suptitle(f"Samples From an EKO Solution")

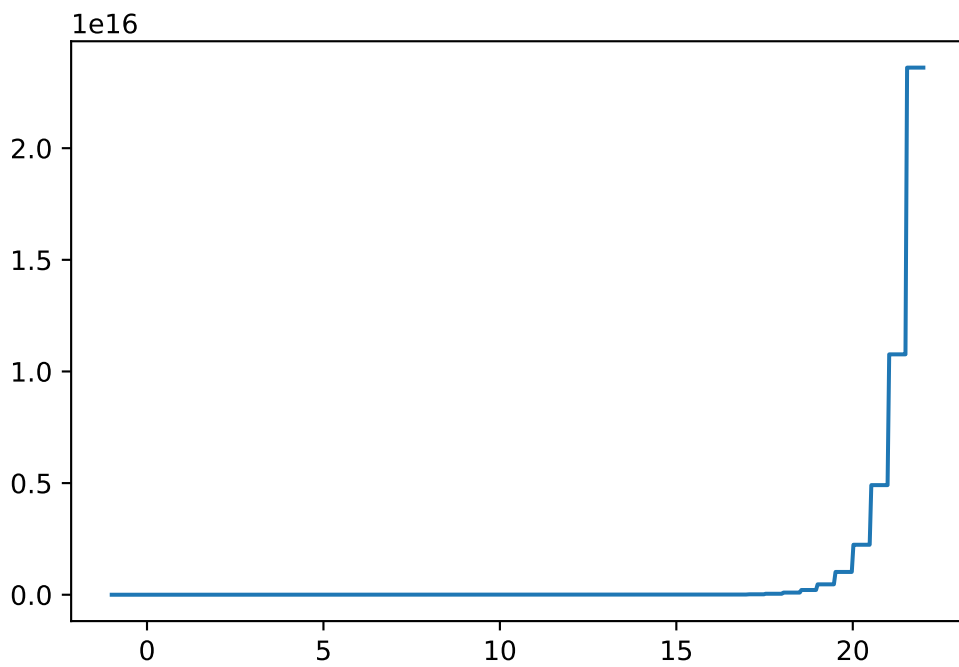
plt.show()

```

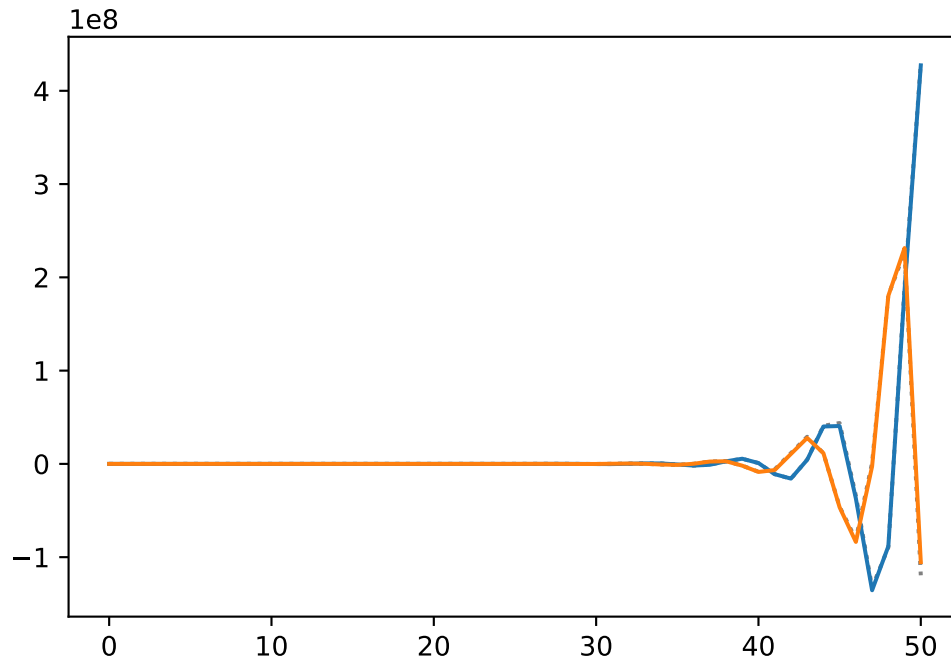
Samples From an EK0 Solution



```
[12]: x = np.linspace(-1, 22, 500)
plt.plot(x, sol.kalman_posterior.diffusion_model(x))
plt.show()
```



```
[13]: plt.plot(samples.mean(axis=0), color="gray", linestyle="dotted")
plt.plot(means)
plt.show()
```

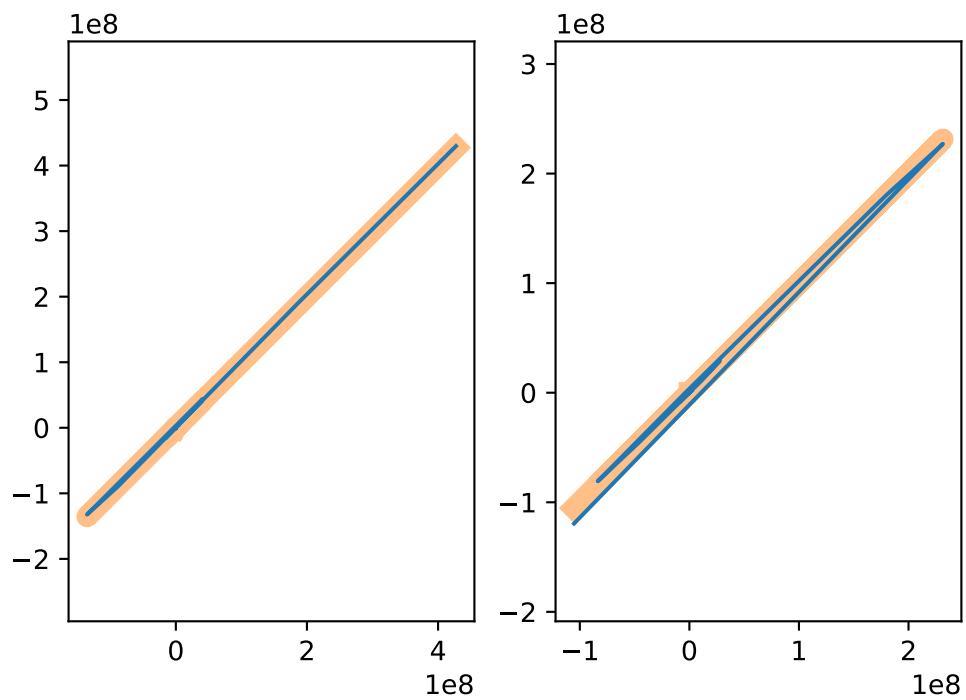


```
[14]: sample_mean = samples.mean(axis=0)
sample_std = samples.std(axis=0)

fig, (ax1, ax2) = plt.subplots(ncols=2)

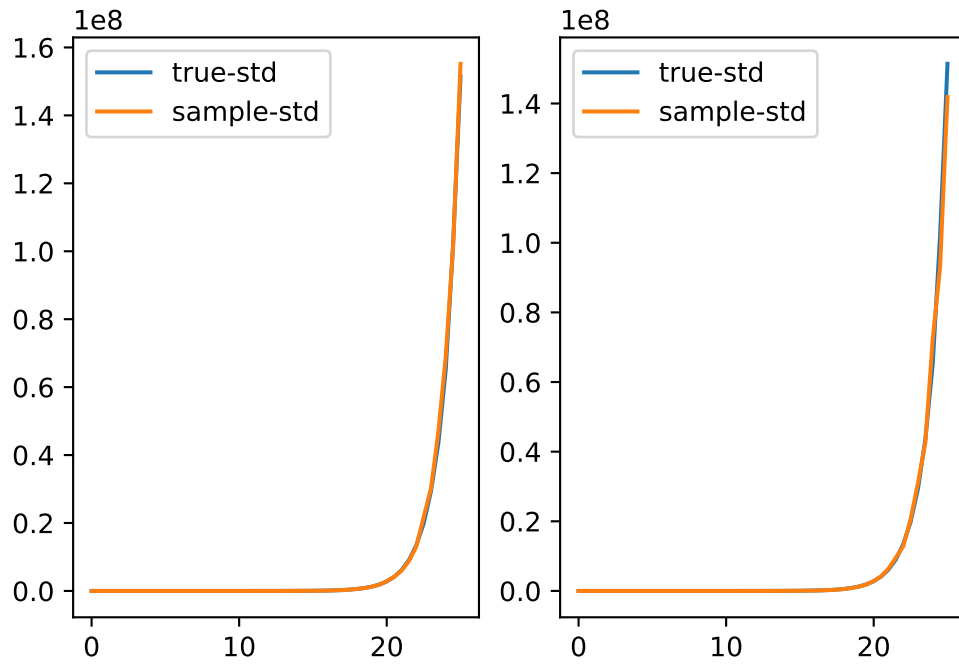
ax1.plot(means[:, 0], sample_mean[:, 0])
ax2.plot(means[:, 1], sample_mean[:, 1])
ax1.plot(means[:, 0], means[:, 0], alpha=0.5, linewidth=8, zorder=0)
ax2.plot(means[:, 1], means[:, 1], alpha=0.5, linewidth=8, zorder=0)

ax1.axis("equal")
ax2.axis("equal")
plt.show()
```



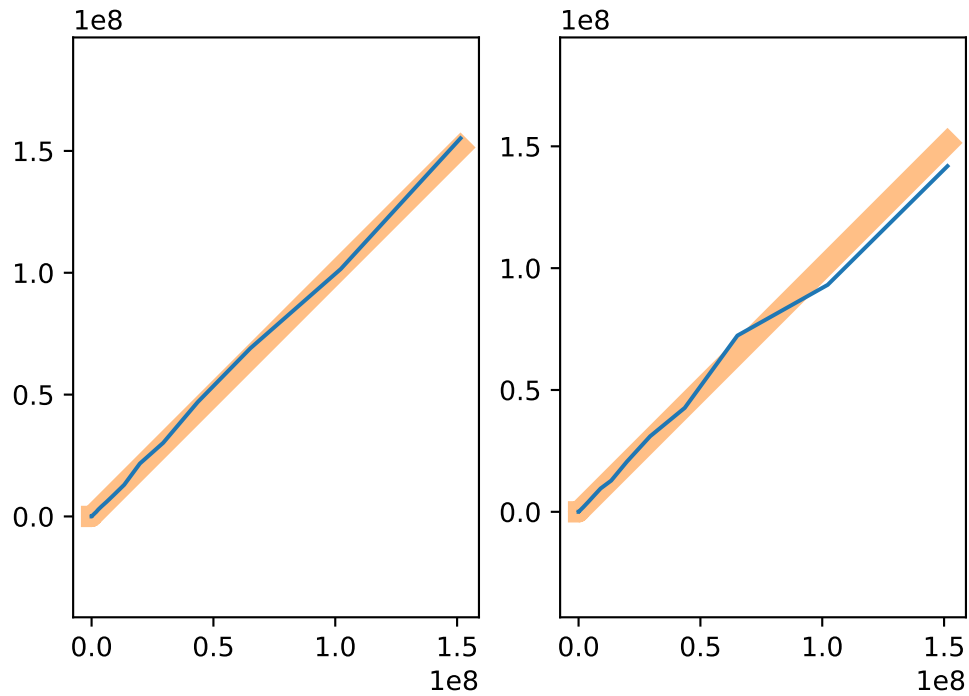
```
[15]: fig, (ax1, ax2) = plt.subplots(ncols=2)

ax1.plot(locations, stds[:, 0], label="true-std")
ax1.plot(locations, sample_std[:, 0], label="sample-std")
ax1.legend()
ax2.plot(locations, stds[:, 1], label="true-std")
ax2.plot(locations, sample_std[:, 1], label="sample-std")
ax2.legend()
plt.show()
```



```
[16]: fig, (ax1, ax2) = plt.subplots(ncols=2)

ax1.plot(stds[:, 0], sample_std[:, 0])
ax2.plot(stds[:, 1], sample_std[:, 1])
ax1.plot(stds[:, 0], stds[:, 0], alpha=0.5, linewidth=8, zorder=0)
ax2.plot(stds[:, 1], stds[:, 1], alpha=0.5, linewidth=8, zorder=0)
ax1.axis("equal")
ax2.axis("equal")
plt.show()
```



```
[ ]:
```

```
[17]: from probnum import diffeq, filtsmooth, randvars, randprocs, problems
import numpy as np

import matplotlib.pyplot as plt
```

```
[18]: def f(t, y):
    y1, y2 = y
    return np.array([-0.1*y1+2*y2, -2*y1-0.1*y2])

def df(t, y):
    y1, y2 = y
    return np.array([[ -0.1, 2], [-2, -0.1]])

t0 = 0.0
tmax = 25.0
y0 = np.array([2, 0])

ivp = problems.InitialValueProblem(t0=t0, tmax=tmax, y0=y0, f=f, df=df)
```

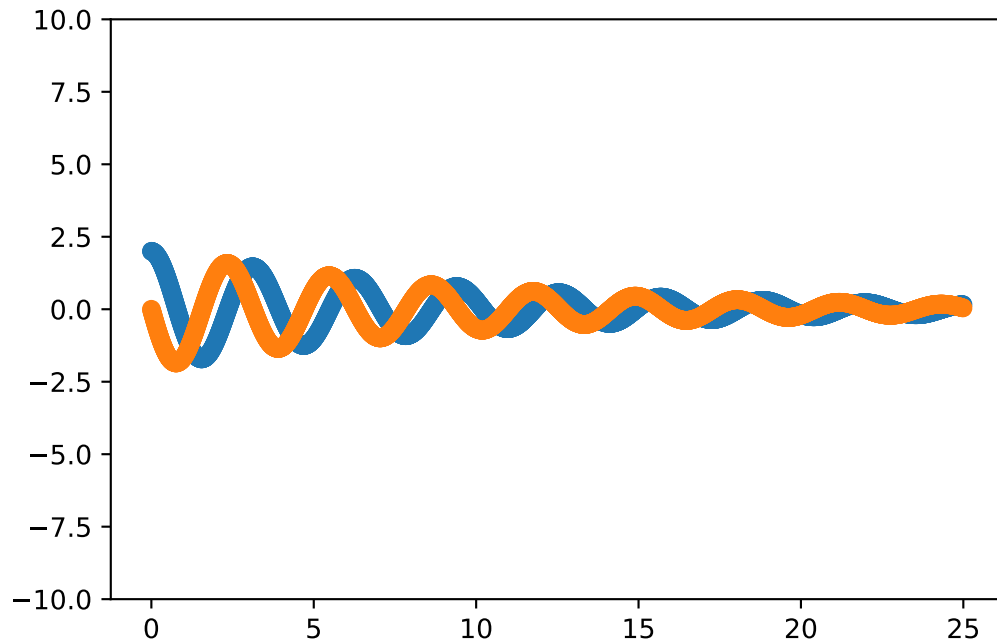
```
[19]: prior_process = randprocs.markov.integrator.IntegratedWienerProcess(
        initarg=ivp.t0,
        num_derivatives=4,
        wiener_process_dimension=ivp.dimension,
        forward_implementation="sqrt",
        backward_implementation="sqrt",
    )

[20]: diffmodel = randprocs.markov.continuous.PiecewiseConstantDiffusion(t0=t0)
rk_init = diffeq.odefilter.initialization_routines.RungeKuttaInitialization()
ode_residual = diffeq.odefilter.information_operators.ODEResidual(4, ivp.
    ↳dimension)
ek1 = diffeq.odefilter.approx_strategies.EK1()
firststep = diffeq.stepsize.propose_firststep(ivp)
steprule = diffeq.stepsize.AdaptiveSteps(firststep=firststep, atol=1e-3,
    ↳rtol=1e-5)

solver = diffeq.odefilter.ODEFilter(
    steprule,
    prior_process=prior_process,
    information_operator=ode_residual,
    approx_strategy=ek1,
    initialization_routine=rk_init,
    diffusion_model=diffmodel,
    with_smoothing=True,
)

[21]: odesol = solver.solve(ivp=ivp)
evalgrid = np.arange(ivp.t0, ivp.tmax, step=0.01)
sol = odesol(evalgrid)

plt.plot(evalgrid, sol.mean, "o-", linewidth=1)
plt.ylim((-10, 10))
plt.show()
```



[]:

2 2-D damped oscillator (Nonlinear)

```
[22]: def f(t, y):
        y1, y2 = y
        return np.array([-0.1*y1**3+2*y2**3, -2*y1**3-0.1*y2**3])

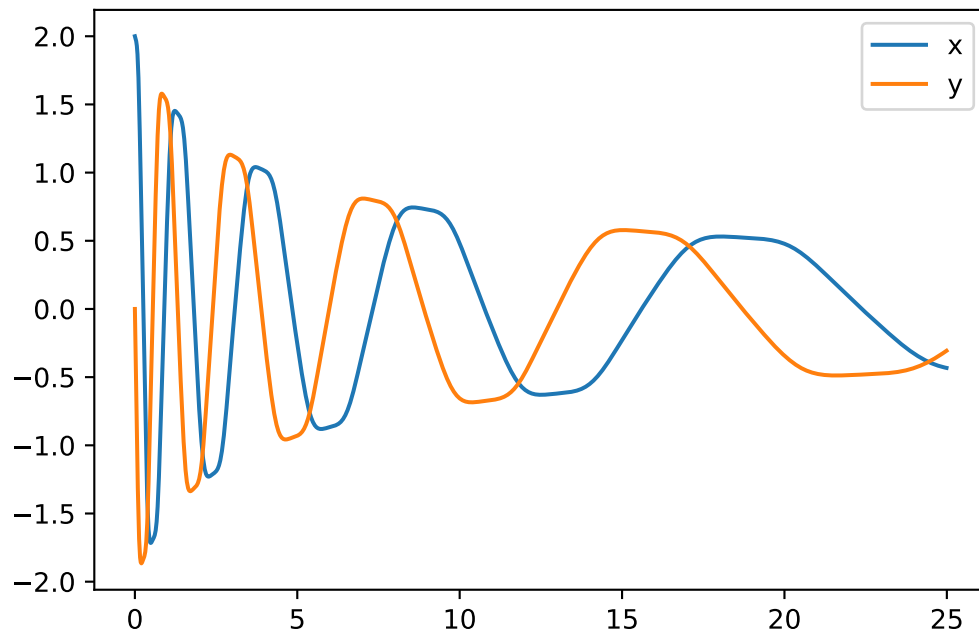
    def df(t, y):
        y1, y2 = y
        return np.array([[-0.3*y1**2, 2*y2**2], [-2*y1**2, -0.1*y2**2]])

    t0 = 0.0
    tmax = 25.0
    y0 = np.array([2, 0])
    ivp = problems.InitialValueProblem(t0=t0, tmax=tmax, y0=y0, f=f, df=df)

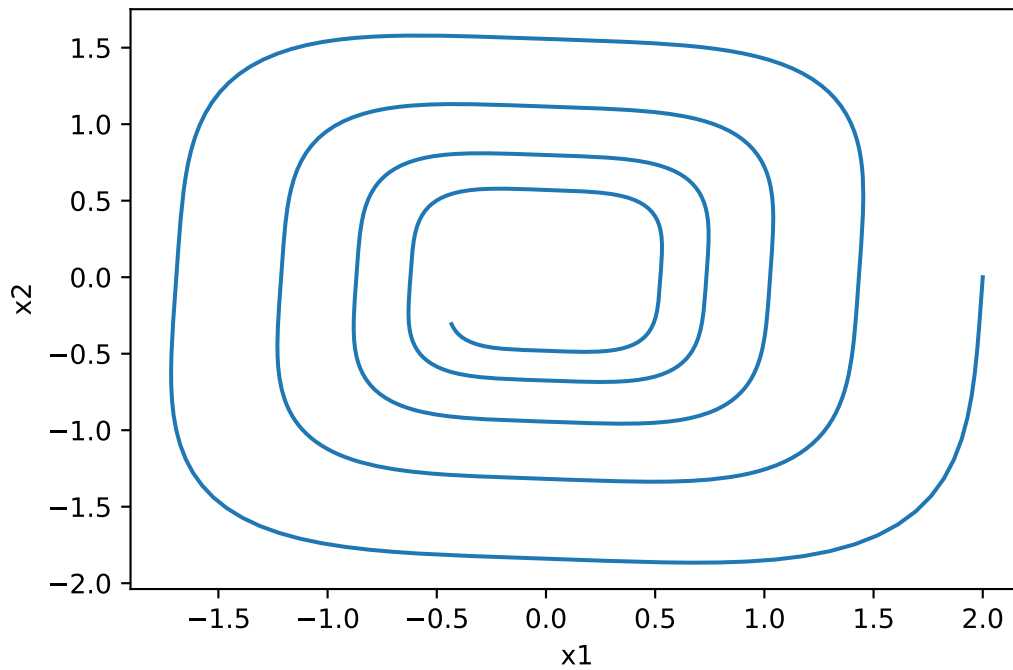
[23]: # IBM(1) with EK0, solving an IVP with adaptive steps works.
    sol = probsolve_ivp(t0=t0, tmax=tmax, y0=y0, f=f, df=df)
    dt=0.01
    evalgrid = np.arange(0, 25, dt)
```

```
means = sol(evalgrid).mean

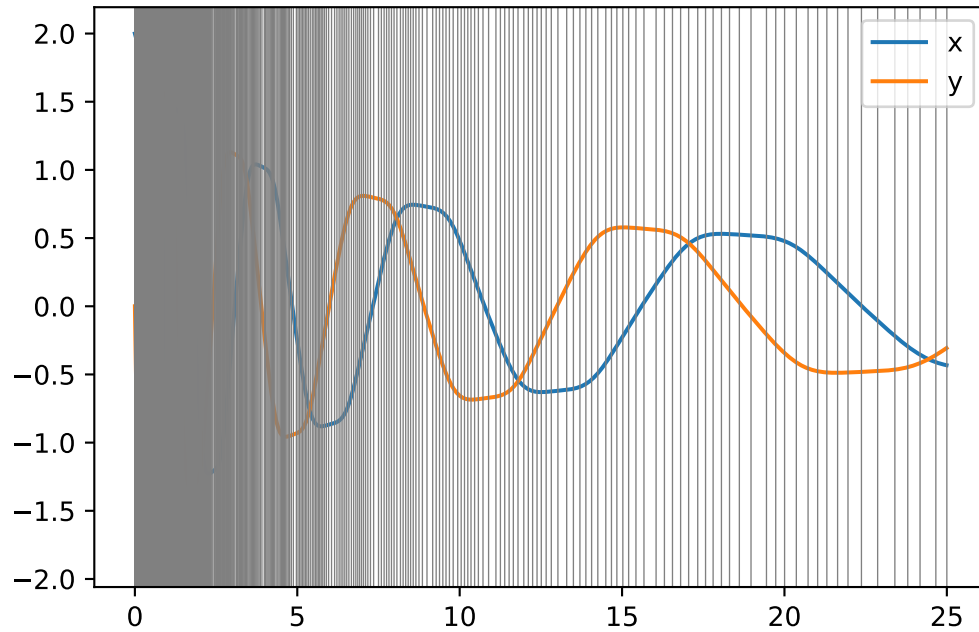
plt.plot(evalgrid, means[:, 0], label="x")
plt.plot(evalgrid, means[:, 1], label="y")
plt.legend(loc="upper right")
plt.show()
```



```
[24]: plt.plot(means[:, 0], means[:, 1])
plt.xlabel("x1")
plt.ylabel("x2")
plt.show()
```



```
[25]: # Let's visualise the individual steps.
plt.plot(evalgrid, means[:, 0], label="x")
plt.plot(evalgrid, means[:, 1], label="y")
for t in sol.locations:
    plt.axvline(t, linewidth=0.5, color="gray")
plt.legend(loc="upper right")
plt.show()
```



```
[26]: algo_orders = [2, 3]
filters = ["EK0", "EK1"]

fig, ax = plt.subplots(
    nrows=len(algo_orders),
    ncols=len(filters),
    sharex=True,
    sharey=True,
    figsize=(12, 12),
)

evalgrid = np.linspace(t0, tmax, 100)

for idx in range(len(algo_orders)): # algo_orders in rows
    for jdx in range(len(filters)): # filters in cols
        sol = probsolve_ivp(
            f,
            t0,
            tmax,
            y0,
            df=df,
            algo_order=algo_orders[idx],
            method=filters[jdx],
        )
```

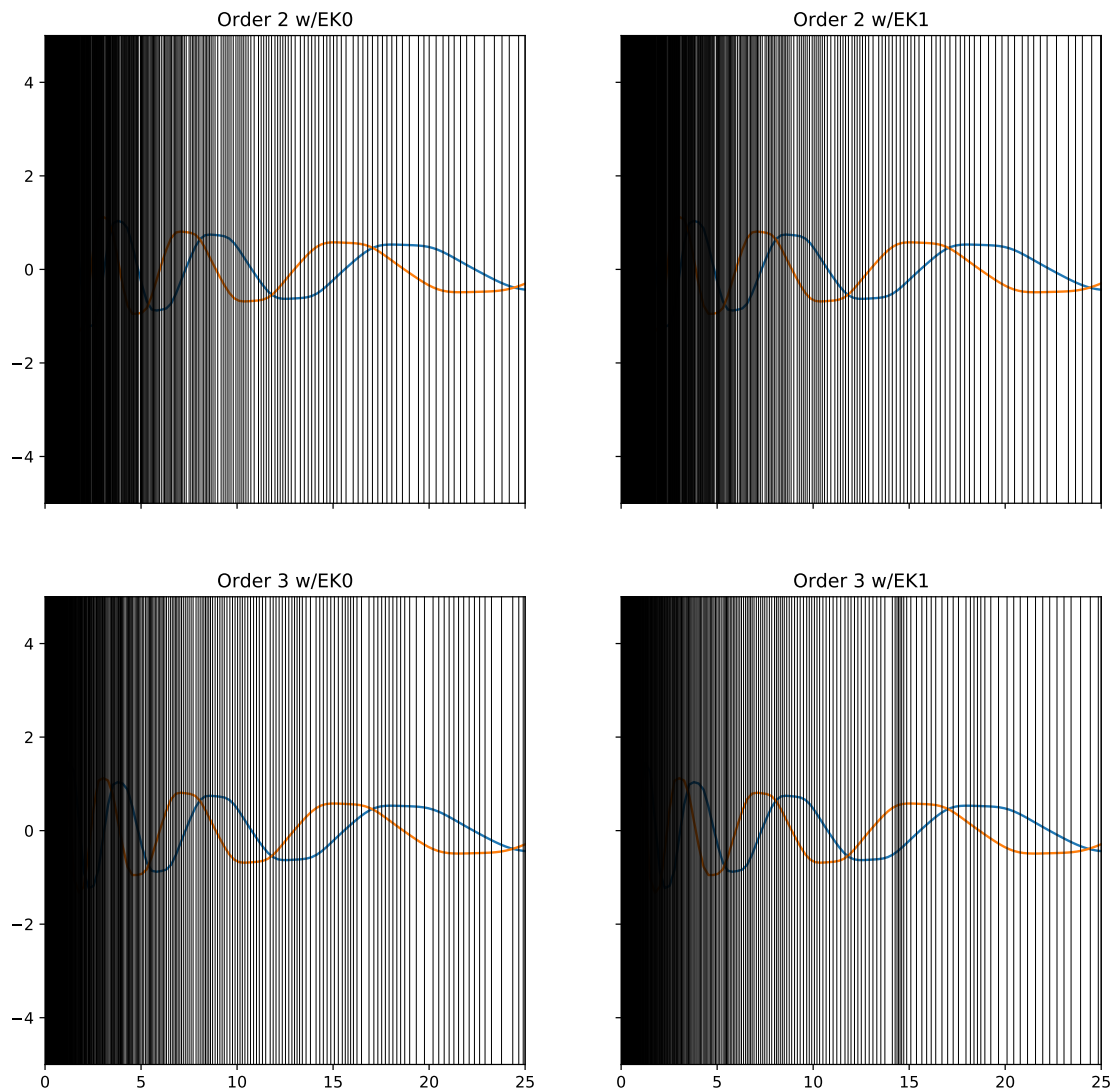
```

solution = sol(evalgrid)
ts, means, stds = evalgrid, solution.mean, solution.std
ax[idx][jdx].plot(ts, means)

for t in sol.locations:
    ax[idx][jdx].axvline(t, linewidth=0.2, color="black")
ax[idx][jdx].set_title(f"Order {algo_orders[idx]} w/{filters[jdx]}")
ax[idx][jdx].set_xlim((t0, tmax))
ax[idx][jdx].set_ylim((-5, 5))

plt.show()

```



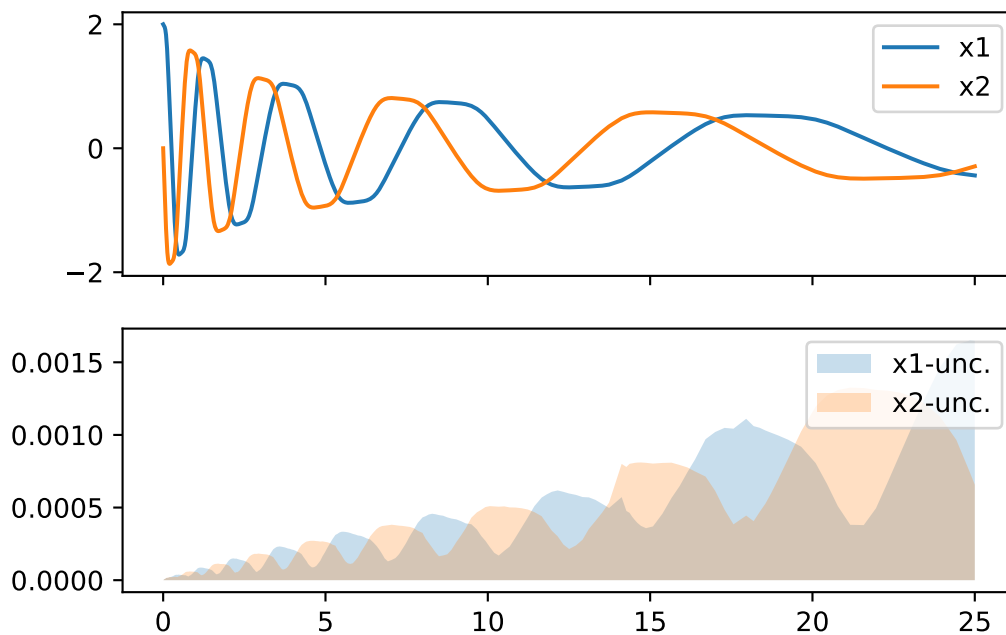
```
[27]: """
sol = probsolve_ivp(
    f, t0, tmax, y0, df=df, step=0.01, adaptive=False, diffusion_model="dynamic"
)
"""
```

```
[27]: '\nsol = probsolve_ivp(\n    f, t0, tmax, y0, df=df, step=0.01, adaptive=False,\n    diffusion_model="dynamic"\n)\n'
```

```
[28]: means, stds = sol.states.mean, sol.states.std

fig, (ax1, ax2) = plt.subplots(nrows=2, ncols=1, sharex=True)
ax1.plot(sol.locations, means[:, 0], label="x1")
ax1.plot(sol.locations, means[:, 1], label="x2")
ax1.legend()
ax2.fill_between(sol.locations, stds[:, 0], alpha=0.25, label="x1-unc.")
ax2.fill_between(sol.locations, stds[:, 1], alpha=0.25, label="x2-unc.")
ax2.legend()
fig.suptitle("EK0 Solution")
plt.show()
```

EK0 Solution



```
[29]: """
sol = probsolve_ivp(
```

```

    f,
    t0,
    tmax,
    y0,
    df=df,
    step=0.5,
    algo_order=1,
    adaptive=False,
    diffusion_model="dynamic",
)
"""

```

```

[29]: '\nsol = probsolve_ivp(\n    f,\n    t0,\n    tmax,\n    y0,\n    df=df,\n    step=0.5,\n    algo_order=1,\n    adaptive=False,\n    diffusion_model="dynamic",\n)\n'

```

```

[30]: import numpy as np

from probnum.diffeq import probsolve_ivp

rng = np.random.default_rng(seed=123)

num_samples = 150
locations = np.arange(0.0, 22.1, 0.05)
locations = sol.locations
samples = sol.sample(rng=rng, t=locations, size=num_samples)

solution = sol(locations)
means = solution.mean
stds = solution.std

```

```

[31]: fig, (ax1) = plt.subplots(nrows=1, ncols=1, sharex=True)
for sample in samples[::5]:
    ax1.plot(locations, sample[:, 0], ":", color="C0", linewidth=0.5, alpha=1)
    ax1.plot(locations, sample[:, 1], ":", color="C1", linewidth=0.5, alpha=1)
ax1.plot(locations, means[:, 0], label="x1", color="k")
ax1.plot(locations, means[:, 1], label="x2", color="k")
ax1.fill_between(
    locations,
    means[:, 0] - 3 * stds[:, 0],
    means[:, 0] + 3 * stds[:, 0],
    color="C0",
    alpha=0.3,
)
ax1.fill_between(
    locations,

```

```

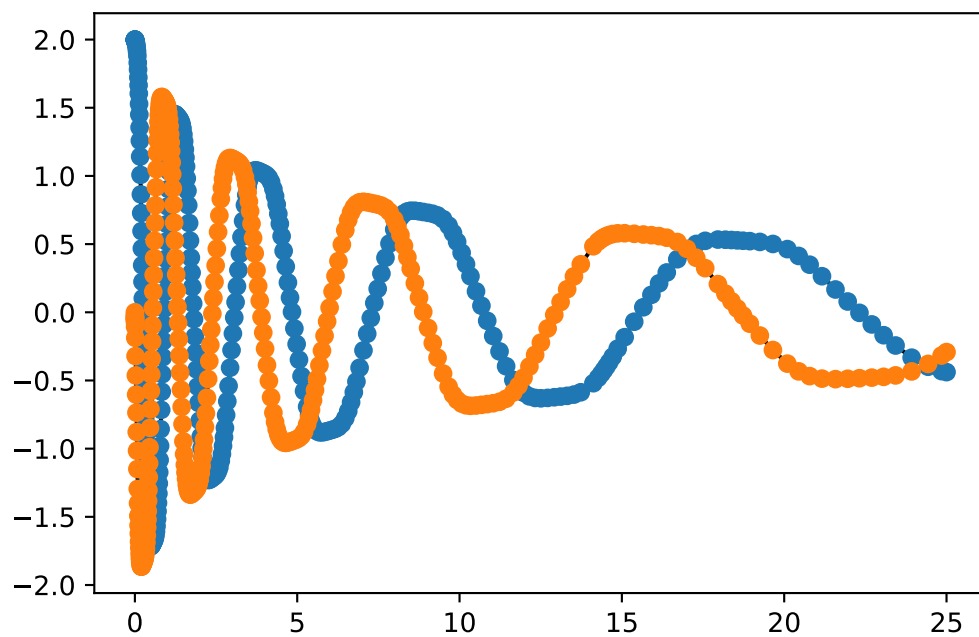
means[:, 1] - 3 * stds[:, 1],
means[:, 1] + 3 * stds[:, 1],
color="C1",
alpha=0.3,
)

ax1.plot(sol.locations, sol.states.mean, "o")
fig.suptitle(f"Samples From an EK0 Solution")

plt.show()

```

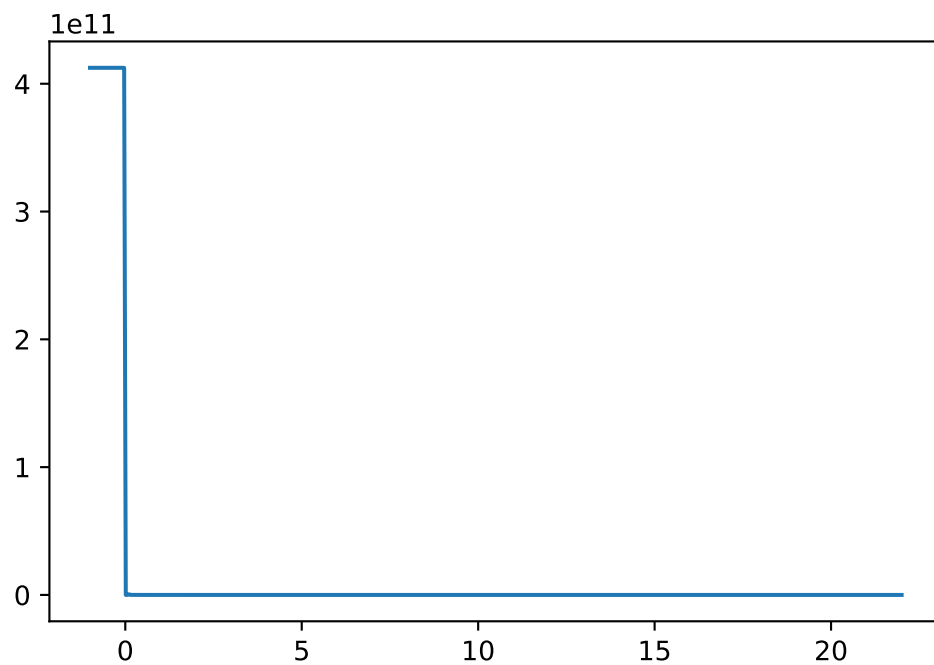
Samples From an EK0 Solution



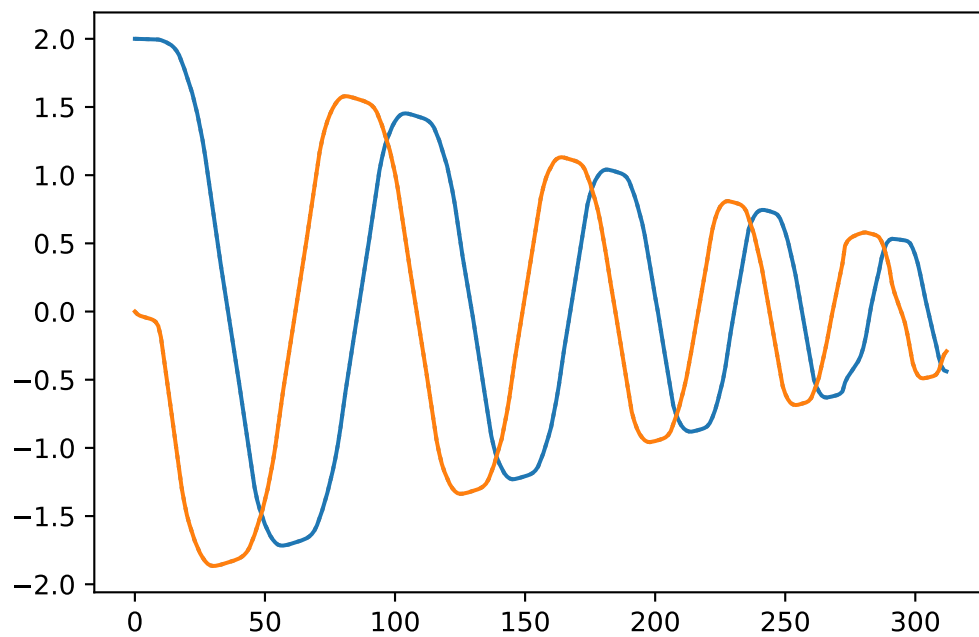
```

[32]: x = np.linspace(-1, 22, 500)
plt.plot(x, sol.kalman_posterior.diffusion_model(x))
plt.show()

```



```
[33]: plt.plot(samples.mean(axis=0), color="gray", linestyle="dotted")  
plt.plot(means)  
plt.show()
```

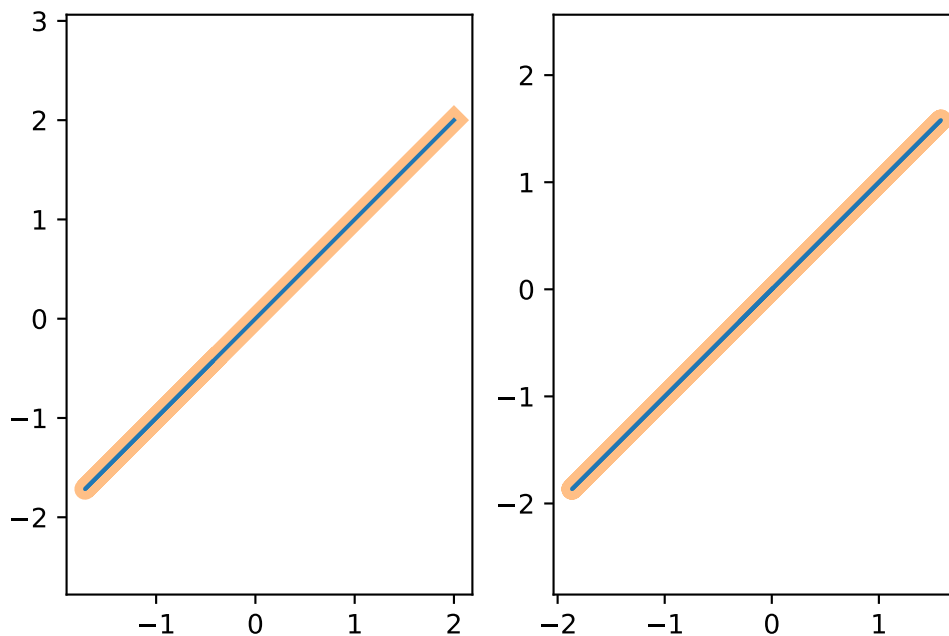


```
[34]: sample_mean = samples.mean(axis=0)
sample_std = samples.std(axis=0)

fig, (ax1, ax2) = plt.subplots(ncols=2)

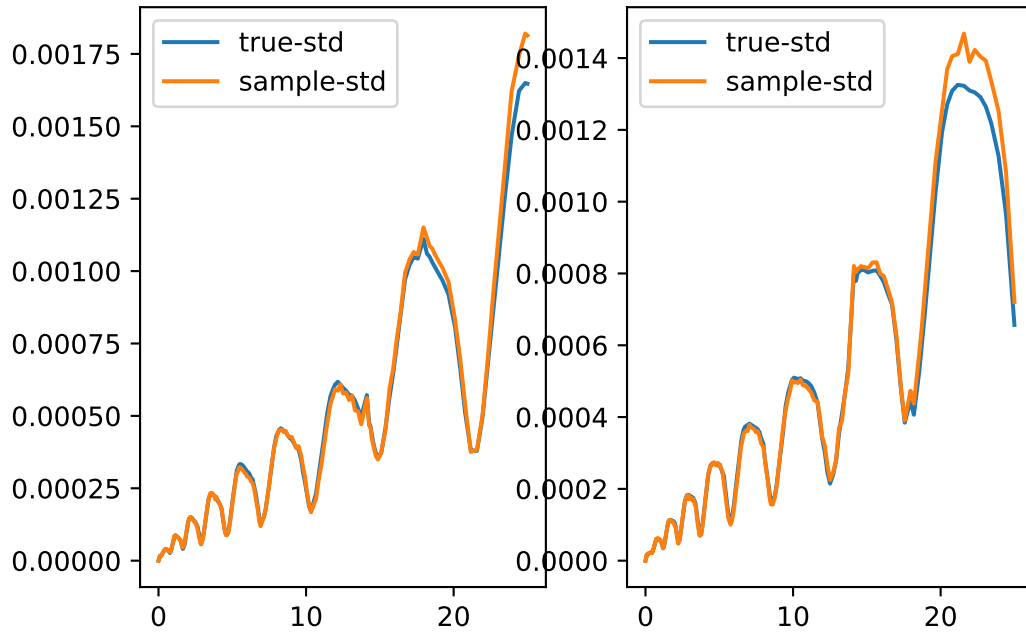
ax1.plot(means[:, 0], sample_mean[:, 0])
ax2.plot(means[:, 1], sample_mean[:, 1])
ax1.plot(means[:, 0], means[:, 0], alpha=0.5, linewidth=8, zorder=0)
ax2.plot(means[:, 1], means[:, 1], alpha=0.5, linewidth=8, zorder=0)

ax1.axis("equal")
ax2.axis("equal")
plt.show()
```



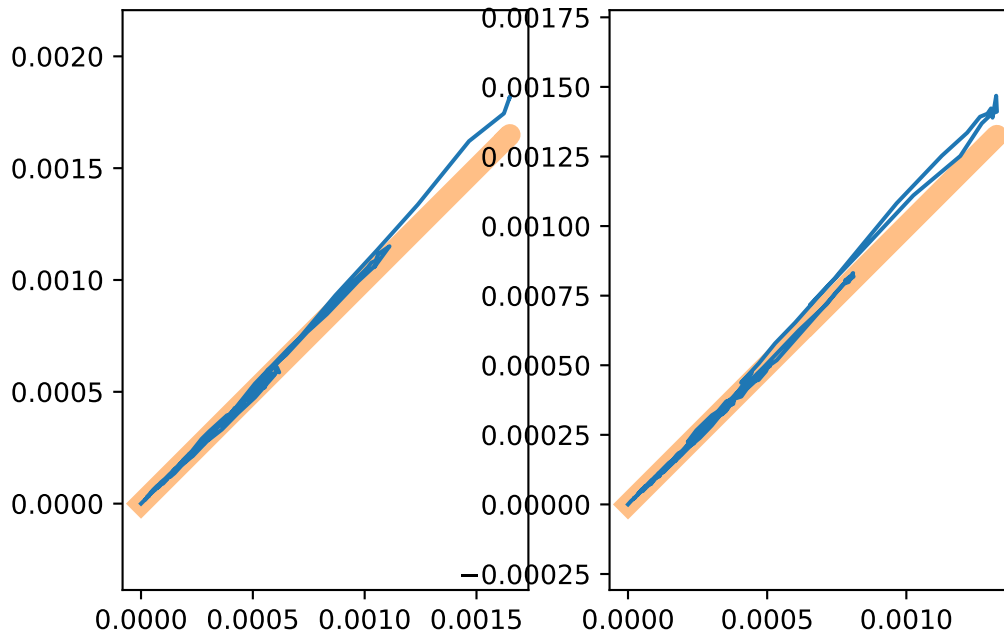
```
[35]: fig, (ax1, ax2) = plt.subplots(ncols=2)

ax1.plot(locations, stds[:, 0], label="true-std")
ax1.plot(locations, sample_std[:, 0], label="sample-std")
ax1.legend()
ax2.plot(locations, stds[:, 1], label="true-std")
ax2.plot(locations, sample_std[:, 1], label="sample-std")
ax2.legend()
plt.show()
```



```
[36]: fig, (ax1, ax2) = plt.subplots(ncols=2)

ax1.plot(stds[:, 0], sample_std[:, 0])
ax2.plot(stds[:, 1], sample_std[:, 1])
ax1.plot(stds[:, 0], stds[:, 0], alpha=0.5, linewidth=8, zorder=0)
ax2.plot(stds[:, 1], stds[:, 1], alpha=0.5, linewidth=8, zorder=0)
ax1.axis("equal")
ax2.axis("equal")
plt.show()
```



```
[37]: from probnum import diffeq, filtsmooth, randvars, randprocs, problems
import numpy as np

import matplotlib.pyplot as plt

def f(t, y):
    y1, y2 = y
    return np.array([-0.1*y1**3+2*y2**3, -2*y1**3-0.1*y2**3])

def df(t, y):
    y1, y2 = y
    return np.array([[-0.3*y1**2, 2*y2**2], [-2*y1**2, -0.1*y2**2]])

t0 = 0.0
tmax = 25.0
y0 = np.array([2, 0])
ivp = problems.InitialValueProblem(t0=t0, tmax=tmax, y0=y0, f=f, df=df)

prior_process = randprocs.markov.integrator.IntegratedWienerProcess(
    initarg=ivp.t0,
    num_derivatives=4,
    wiener_process_dimension=ivp.dimension,
    forward_implementation="sqrt",
```

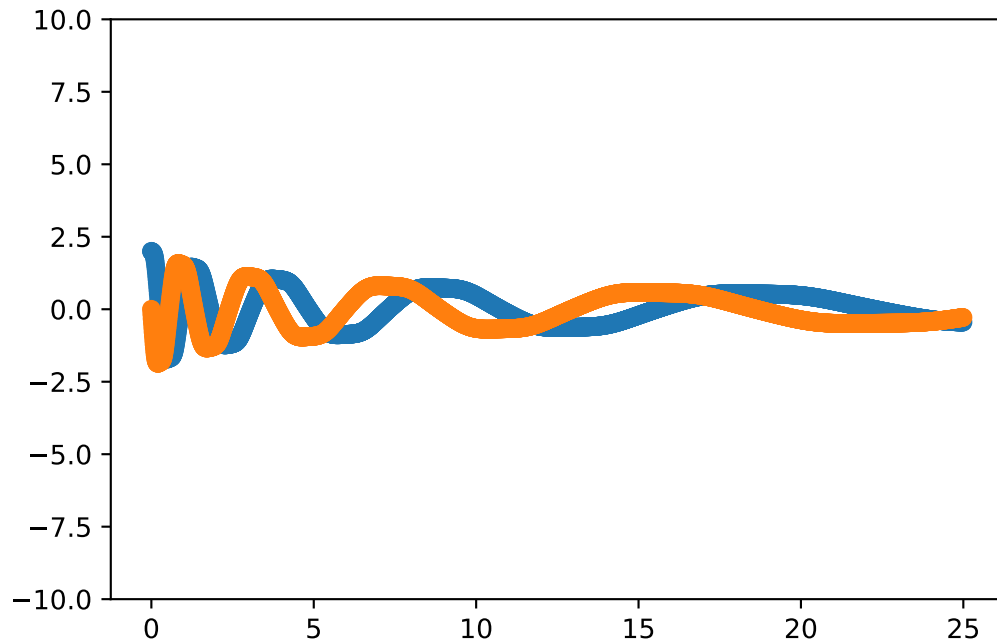
```
backward_implementation="sqrt",
)
```

```
[38]: diffmodel = randprocs.markov.continuous.PiecewiseConstantDiffusion(t0=t0)
rk_init = diffeq.odefilter.initialization_routines.RungeKuttaInitialization()
ode_residual = diffeq.odefilter.information_operators.ODEResidual(4, ivp.
    ↳ dimension)
ek1 = diffeq.odefilter.approx_strategies.EK1()
firststep = diffeq.stepsize.propose_firststep(ivp)
steprule = diffeq.stepsize.AdaptiveSteps(firststep=firststep, atol=1e-3,
    ↳ rtol=1e-5)

solver = diffeq.odefilter.ODEFilter(
    steprule,
    prior_process=prior_process,
    information_operator=ode_residual,
    approx_strategy=ek1,
    initialization_routine=rk_init,
    diffusion_model=diffmodel,
    with_smoothing=True,
)

odesol = solver.solve(ivp=ivp)
evalgrid = np.arange(ivp.t0, ivp.tmax, step=0.01)
sol = odesol(evalgrid)

plt.plot(evalgrid, sol.mean, "o-", linewidth=1)
plt.ylim((-10, 10))
plt.show()
```



[]:

3 Brusselator

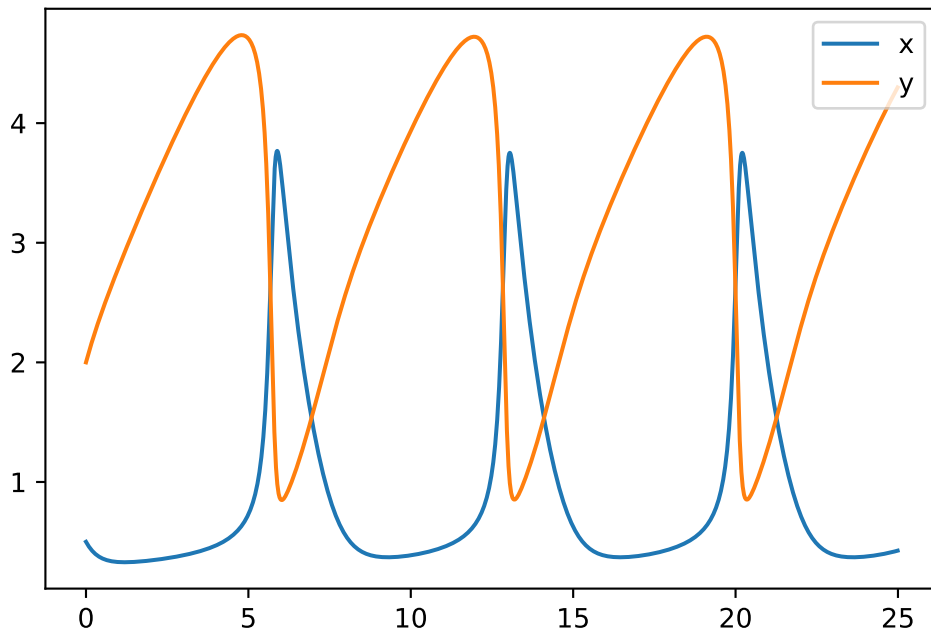
```
[39]: def f(t, y):
        y1, y2 = y
        return np.array([1-4*y1+y1**2*y2, 3*y1-y1**2*y2])

    def df(t, y):
        y1, y2 = y
        return np.array([[-4+2*y1*y2, y1**2], [3-2*y1*y2, -y1**2]])

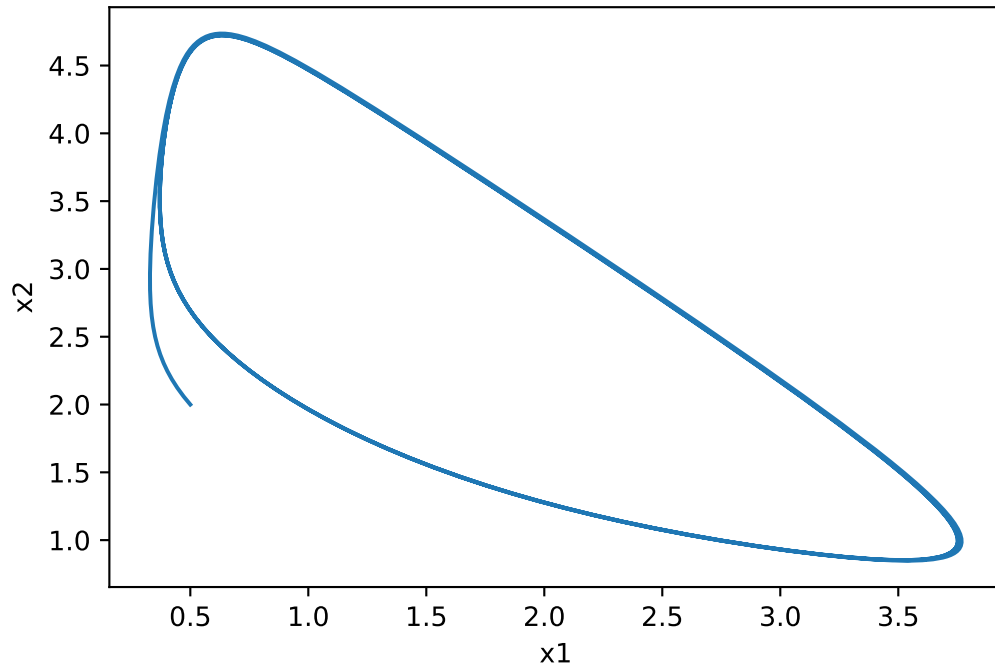
    t0 = 0.0
    tmax = 25.0
    y0 = np.array([0.5, 2])
```

```
[40]: # IBM(1) with EK0, solving an IVP with adaptive steps works.
    sol = probsolve_ivp(f, t0, tmax, y0, df=df, algo_order=4, method="EK1")
    dt=0.01
    evalgrid = np.arange(0, 25, dt)
    means = sol(evalgrid).mean
```

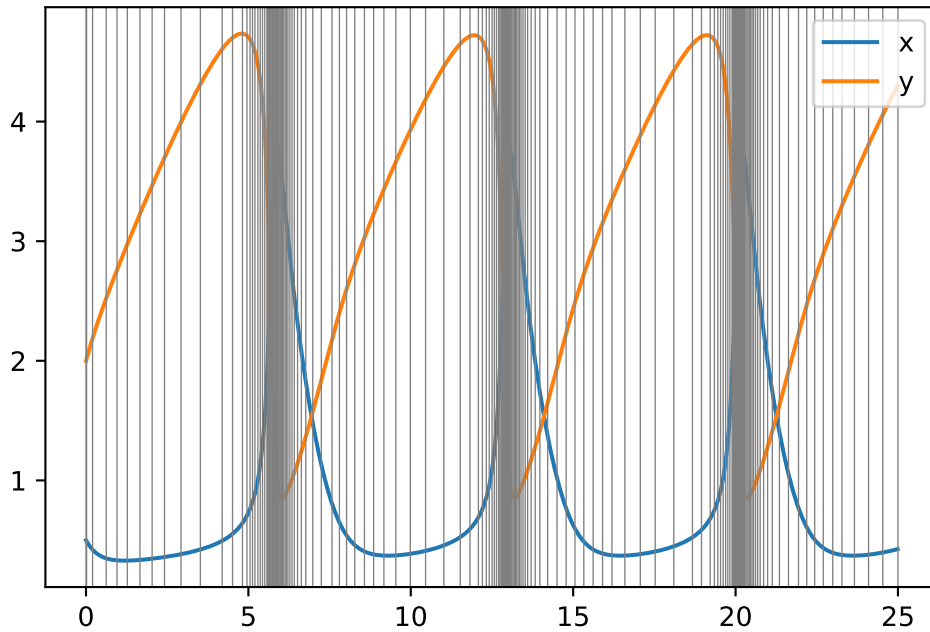
```
plt.plot(evalgrid, means[:, 0], label="x")
plt.plot(evalgrid, means[:, 1], label="y")
plt.legend(loc="upper right")
plt.show()
```



```
[41]: plt.plot(means[:, 0], means[:, 1])
plt.xlabel("x1")
plt.ylabel("x2")
plt.show()
```



```
[42]: # Let's visualise the individual steps.
plt.plot(evalgrid, means[:, 0], label="x")
plt.plot(evalgrid, means[:, 1], label="y")
for t in sol.locations:
    plt.axvline(t, linewidth=0.5, color="gray")
plt.legend(loc="upper right")
plt.show()
```



```
[43]: algo_orders = [2, 3]
filters = ["EK0", "EK1"]

fig, ax = plt.subplots(
    nrows=len(algo_orders),
    ncols=len(filters),
    sharex=True,
    sharey=True,
    figsize=(12, 12),
)

evalgrid = np.linspace(t0, tmax, 100)

for idx in range(len(algo_orders)): # algo_orders in rows
    for jdx in range(len(filters)): # filters in cols
        sol = probsolve_ivp(
            f,
            t0,
            tmax,
            y0,
            df=df,
            algo_order=algo_orders[idx],
            method=filters[jdx],
        )
```

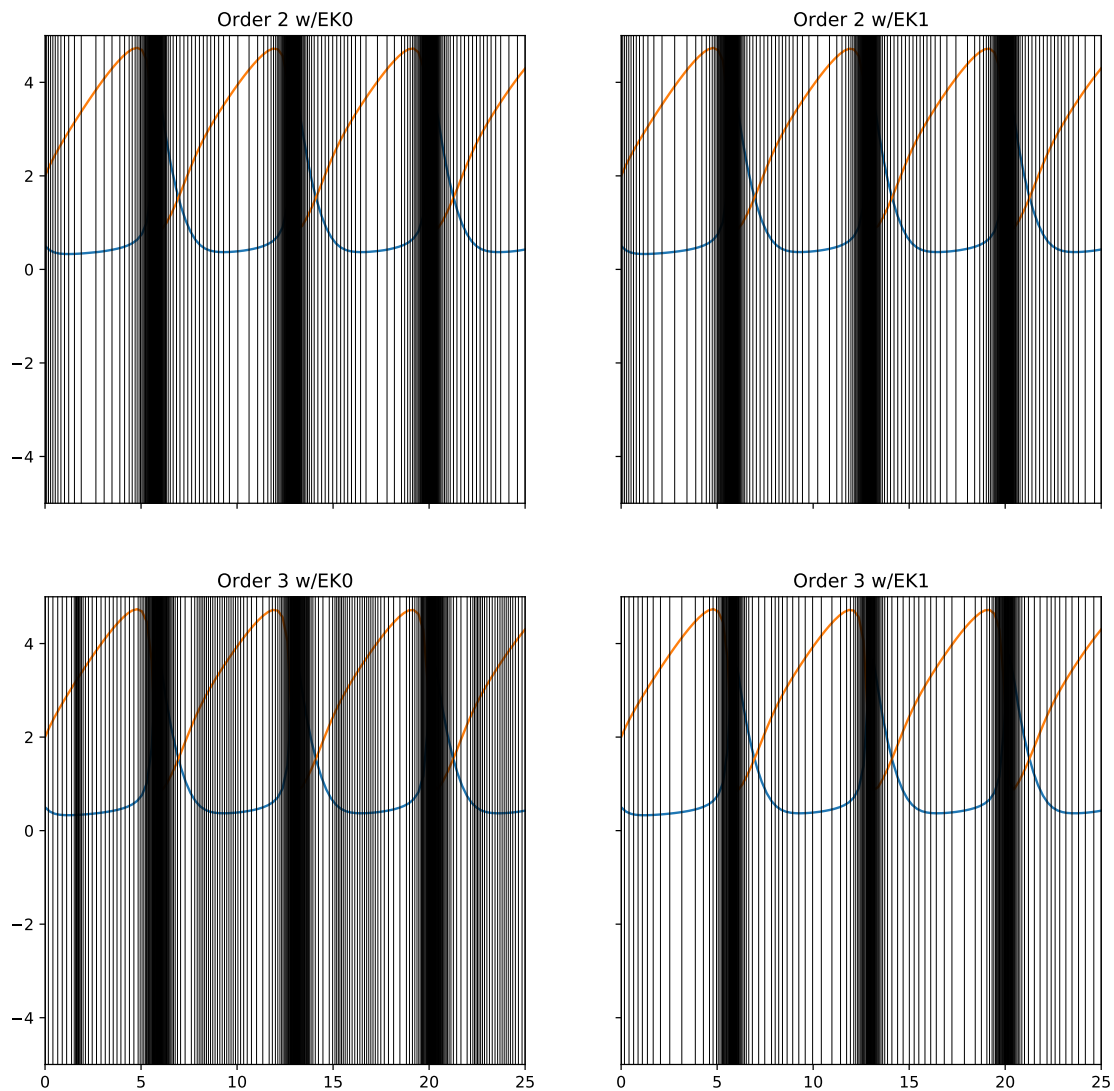
```

solution = sol(evalgrid)
ts, means, stds = evalgrid, solution.mean, solution.std
ax[idx][jdx].plot(ts, means)

for t in sol.locations:
    ax[idx][jdx].axvline(t, linewidth=0.2, color="black")
ax[idx][jdx].set_title(f"Order {algo_orders[idx]} w/{filters[jdx]}")
ax[idx][jdx].set_xlim((t0, tmax))
ax[idx][jdx].set_ylim((-5, 5))

plt.show()

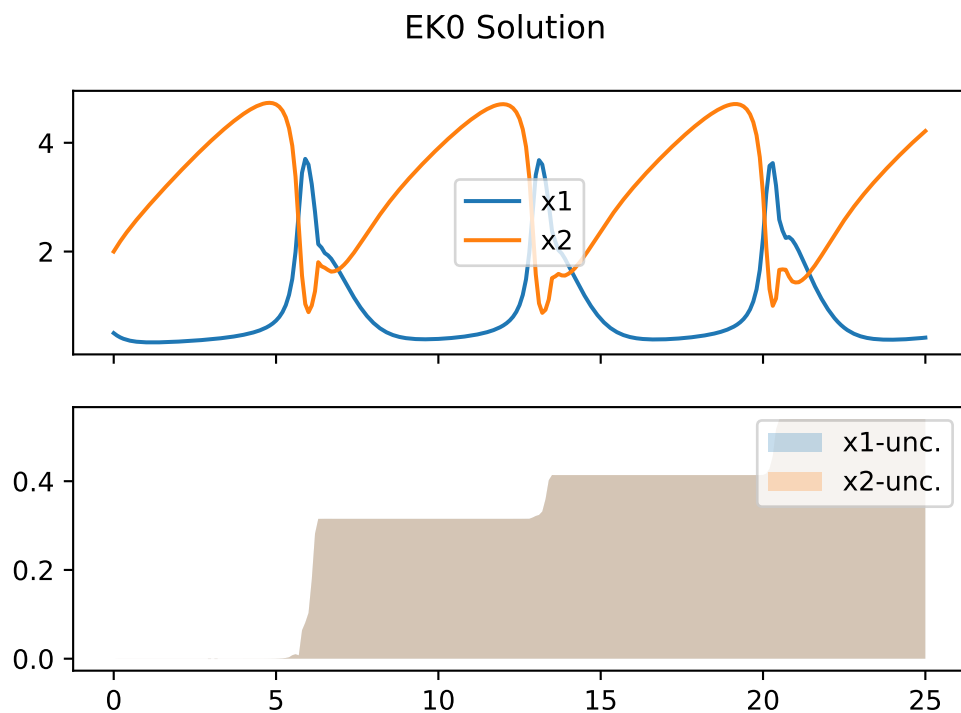
```



```
[44]: sol = probsolve_ivp(
        f, t0, tmax, y0, df=df, step=0.1, adaptive=False, diffusion_model="dynamic"
    )

means, stds = sol.states.mean, sol.states.std

fig, (ax1, ax2) = plt.subplots(nrows=2, ncols=1, sharex=True)
ax1.plot(sol.locations, means[:, 0], label="x1")
ax1.plot(sol.locations, means[:, 1], label="x2")
ax1.legend()
ax2.fill_between(sol.locations, stds[:, 0], alpha=0.25, label="x1-unc.")
ax2.fill_between(sol.locations, stds[:, 1], alpha=0.25, label="x2-unc.")
ax2.legend()
fig.suptitle("EK0 Solution")
plt.show()
```



```
[45]: import numpy as np

from probnum.diffeq import probsolve_ivp

rng = np.random.default_rng(seed=123)
```

```

num_samples = 150
locations = np.arange(0.0, 22.1, 0.05)
locations = sol.locations
samples = sol.sample(rng=rng, t=locations, size=num_samples)

solution = sol(locations)
means = solution.mean
stds = solution.std

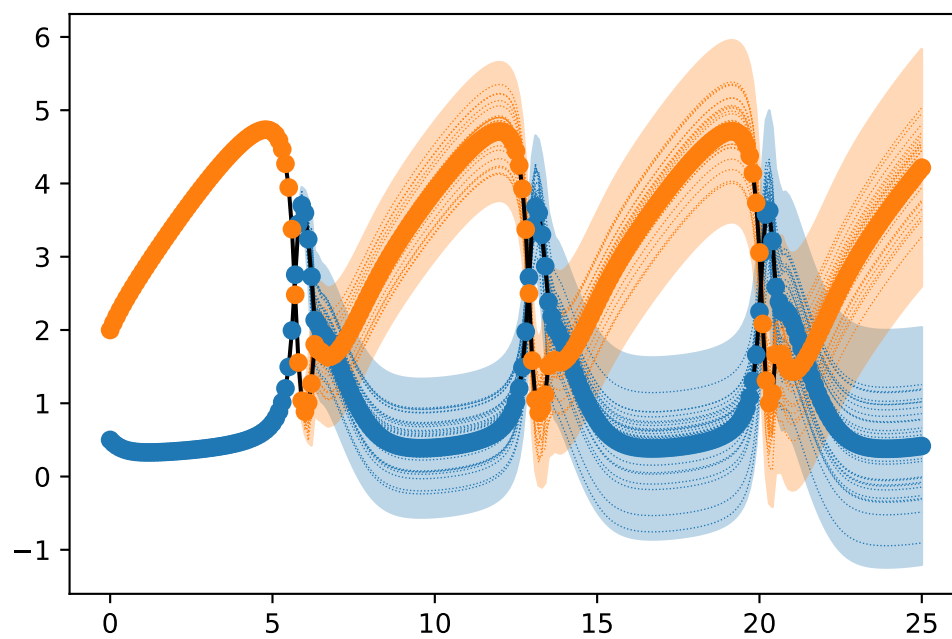
fig, (ax1) = plt.subplots(nrows=1, ncols=1, sharex=True)
for sample in samples[::5]:
    ax1.plot(locations, sample[:, 0], ":", color="C0", linewidth=0.5, alpha=1)
    ax1.plot(locations, sample[:, 1], ":", color="C1", linewidth=0.5, alpha=1)
ax1.plot(locations, means[:, 0], label="x1", color="k")
ax1.plot(locations, means[:, 1], label="x2", color="k")
ax1.fill_between(
    locations,
    means[:, 0] - 3 * stds[:, 0],
    means[:, 0] + 3 * stds[:, 0],
    color="C0",
    alpha=0.3,
)
ax1.fill_between(
    locations,
    means[:, 1] - 3 * stds[:, 1],
    means[:, 1] + 3 * stds[:, 1],
    color="C1",
    alpha=0.3,
)

ax1.plot(sol.locations, sol.states.mean, "o")
fig.suptitle(f"Samples From an EK0 Solution")

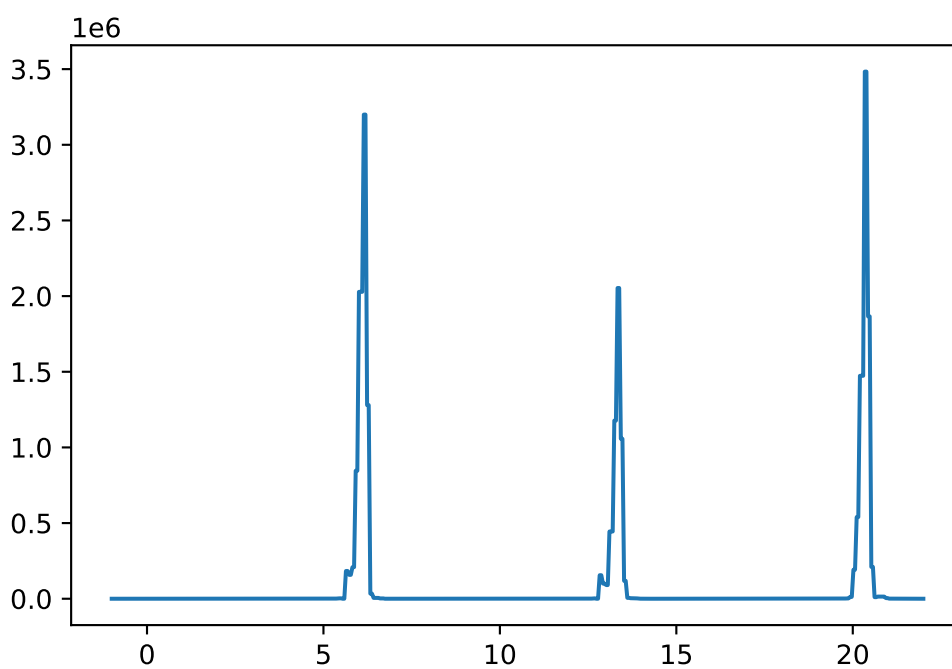
plt.show()

```

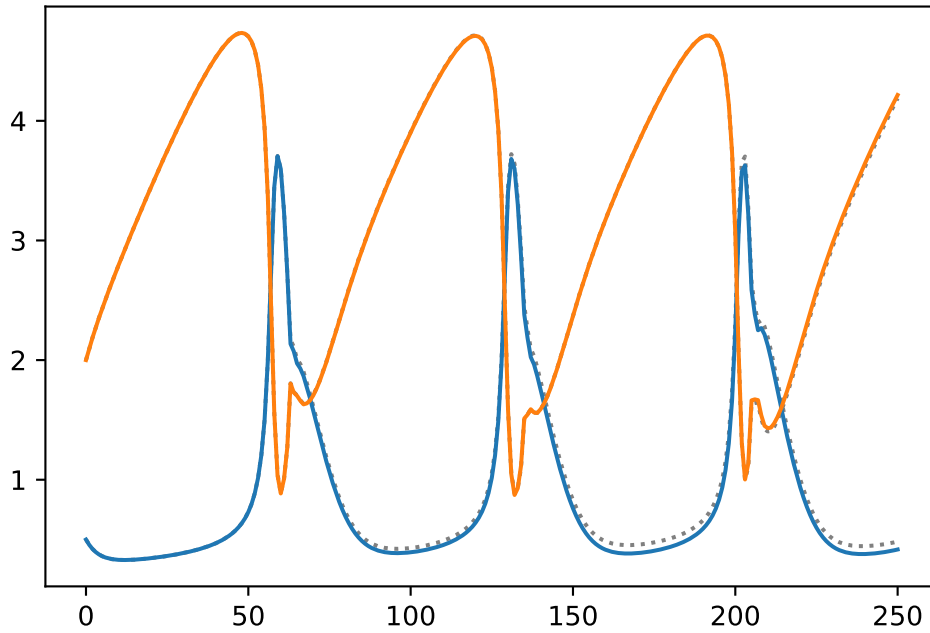
Samples From an EK0 Solution



```
[46]: x = np.linspace(-1, 22, 500)
plt.plot(x, sol.kalman_posterior.diffusion_model(x))
plt.show()
```



```
[47]: plt.plot(samples.mean(axis=0), color="gray", linestyle="dotted")
plt.plot(means)
plt.show()
```

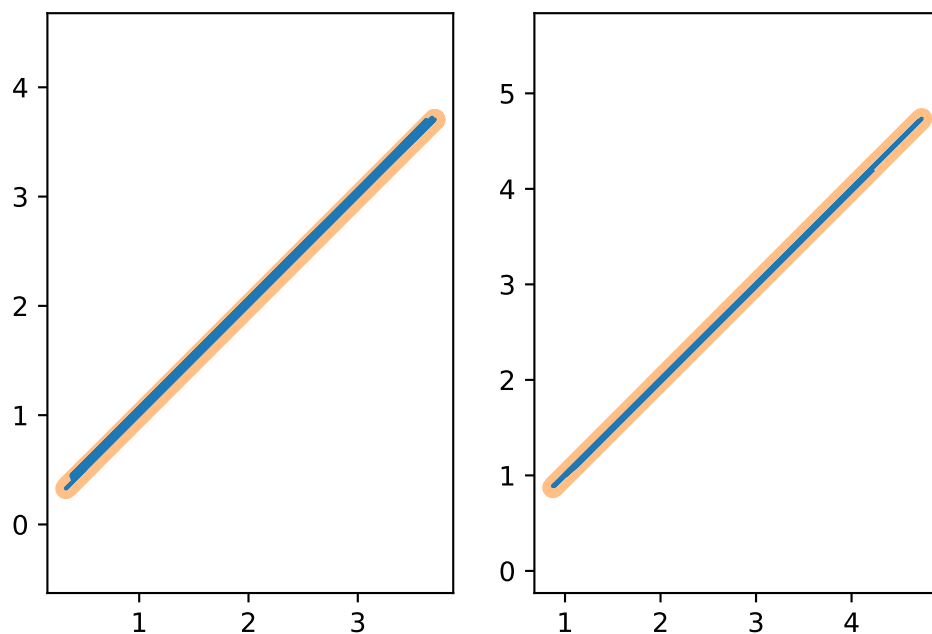


```
[48]: sample_mean = samples.mean(axis=0)
sample_std = samples.std(axis=0)

fig, (ax1, ax2) = plt.subplots(ncols=2)

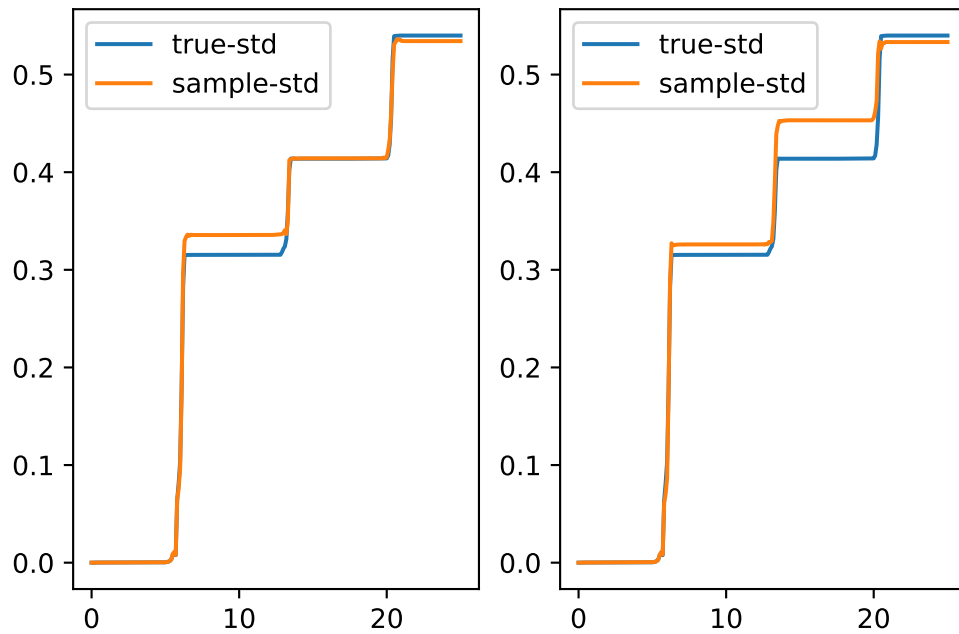
ax1.plot(means[:, 0], sample_mean[:, 0])
ax2.plot(means[:, 1], sample_mean[:, 1])
ax1.plot(means[:, 0], means[:, 0], alpha=0.5, linewidth=8, zorder=0)
ax2.plot(means[:, 1], means[:, 1], alpha=0.5, linewidth=8, zorder=0)

ax1.axis("equal")
ax2.axis("equal")
plt.show()
```



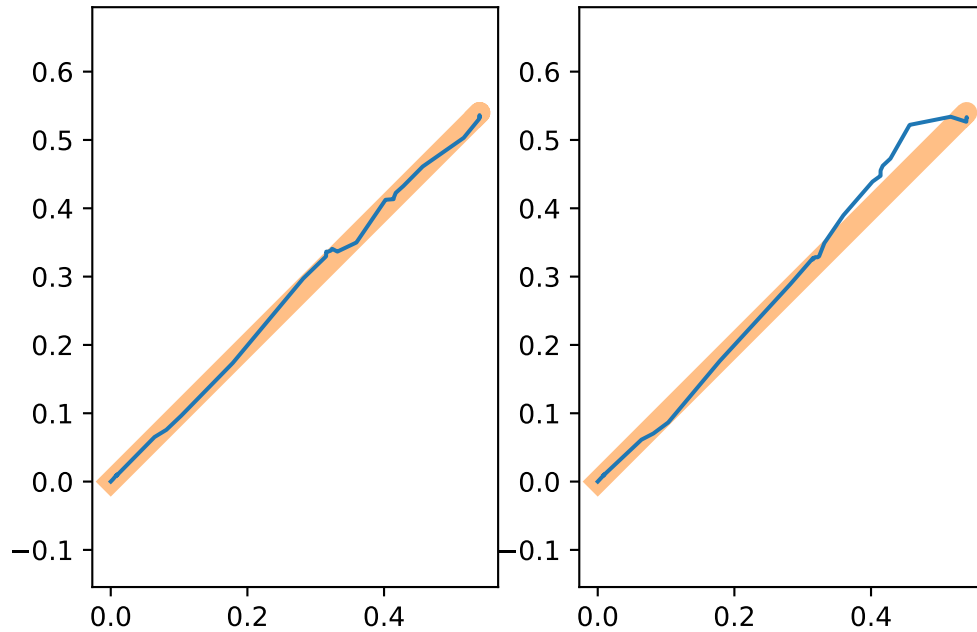
```
[49]: fig, (ax1, ax2) = plt.subplots(ncols=2)

ax1.plot(locations, stds[:, 0], label="true-std")
ax1.plot(locations, sample_std[:, 0], label="sample-std")
ax1.legend()
ax2.plot(locations, stds[:, 1], label="true-std")
ax2.plot(locations, sample_std[:, 1], label="sample-std")
ax2.legend()
plt.show()
```



```
[50]: fig, (ax1, ax2) = plt.subplots(ncols=2)

ax1.plot(stds[:, 0], sample_std[:, 0])
ax2.plot(stds[:, 1], sample_std[:, 1])
ax1.plot(stds[:, 0], stds[:, 0], alpha=0.5, linewidth=8, zorder=0)
ax2.plot(stds[:, 1], stds[:, 1], alpha=0.5, linewidth=8, zorder=0)
ax1.axis("equal")
ax2.axis("equal")
plt.show()
```



4 Three-dimensional linear system

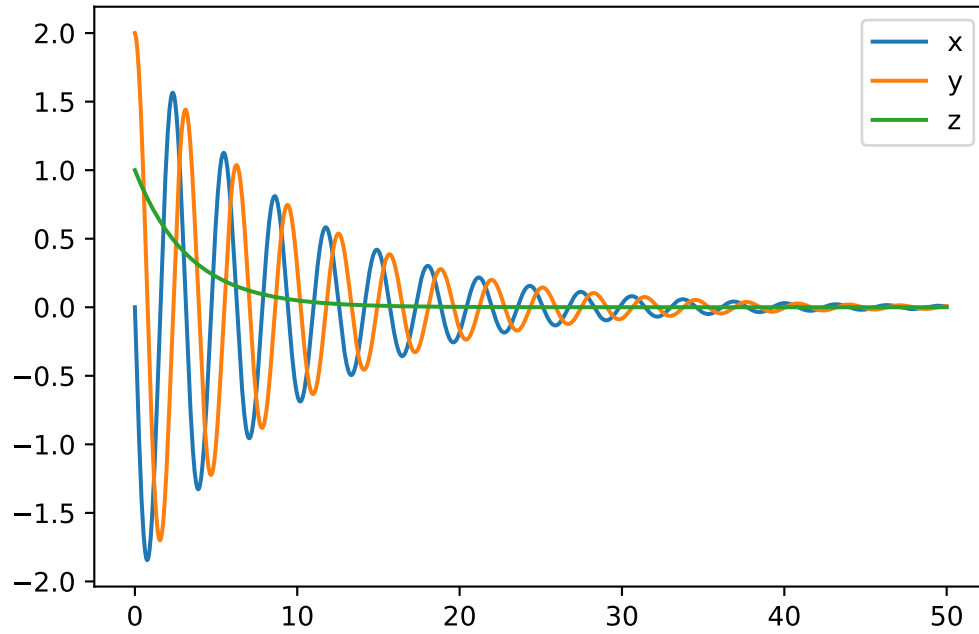
```
[51]: def f(t, y):
    y1, y2, y3 = y
    return np.array([-0.1*y1-2*y2, 2*y1-0.1*y2, -0.3*y3])

def df(t, y):
    y1, y2, y3 = y
    return np.array([[-0.1, 2, 0], [2, -0.1, 0], [0, 0, -0.3]])

t0 = 0.0
tmax = 50.0
y0 = np.array([0, 2, 1])

# IBM(1) with EK0, solving an IVP with adaptive steps works.
sol = probsolve_ivp(f, t0, tmax, y0, df=df, step=0.1, adaptive=False,
    ↪diffusion_model="dynamic")
dt=0.01
evalgrid = np.arange(0, 50, dt)
means = sol(evalgrid).mean
```

```
[52]: plt.plot(evalgrid, means[:, 0], label="x")
plt.plot(evalgrid, means[:, 1], label="y")
plt.plot(evalgrid, means[:, 2], label="z")
plt.legend(loc="upper right")
plt.show()
```



```
[53]: from mpl_toolkits.mplot3d import Axes3D
fig = plt.figure(figsize=(4, 4))
ax = fig.add_subplot(1, 1, 1, projection="3d")
ax.plot(
    means[:, 0],
    means[:, 1],
    means[:, 2]
)
ax.set(title='3D plot', xlabel="$x$", ylabel="$y$", zlabel="$z$")
ax.legend()
fig.show()
```

No handles with labels found to put in legend.

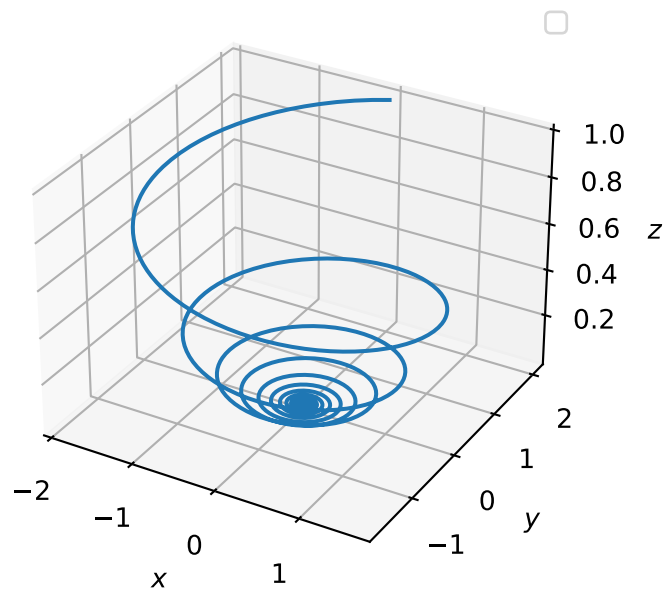
/home/kkumar/anaconda3/lib/python3.7/site-packages/ipykernel_launcher.py:11:

UserWarning: Matplotlib is currently using

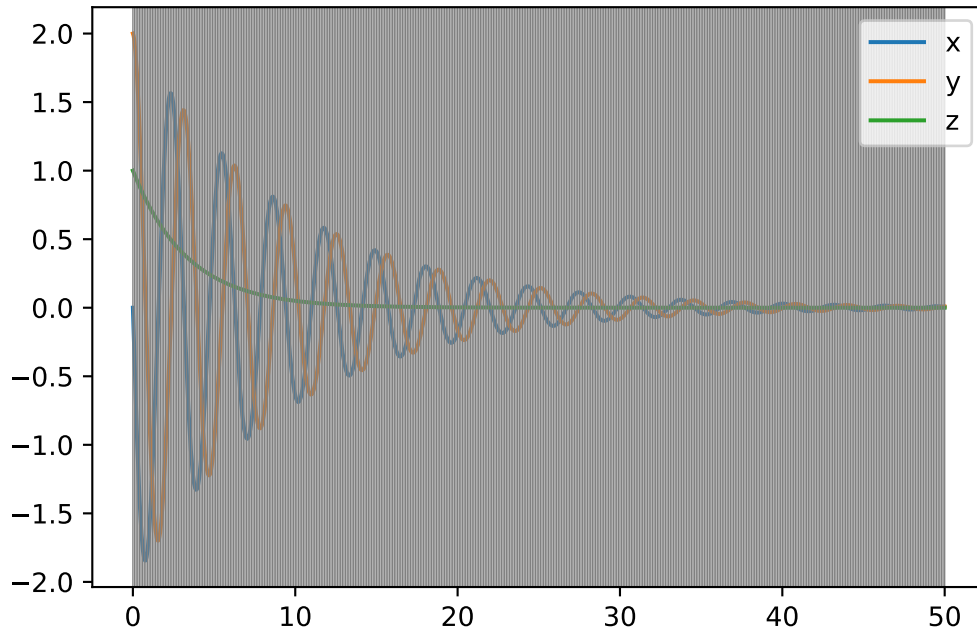
module://ipykernel.pylab.backend_inline, which is a non-GUI backend, so cannot show the figure.

This is added back by InteractiveShellApp.init_path()

3D plot



```
[54]: # Let's visualise the individual steps.
plt.plot(evalgrid, means[:, 0], label="x")
plt.plot(evalgrid, means[:, 1], label="y")
plt.plot(evalgrid, means[:, 2], label="z")
for t in sol.locations:
    plt.axvline(t, linewidth=0.5, color="gray")
plt.legend(loc="upper right")
plt.show()
```

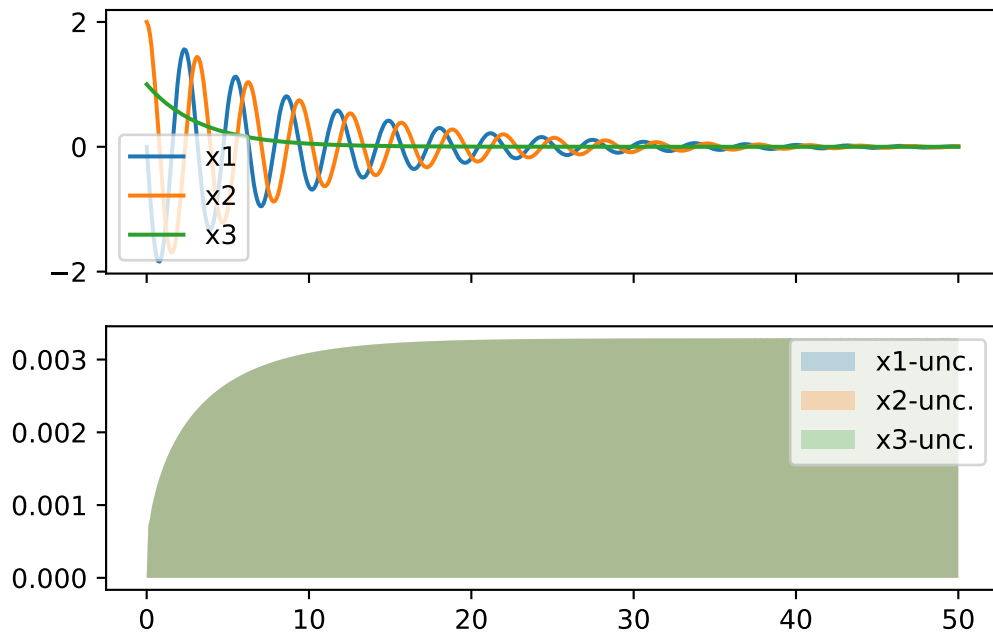


```
[55]: sol = probsolve_ivp(
        f, t0, tmax, y0, df=df, step=0.1, adaptive=False, diffusion_model="dynamic"
    )

means, stds = sol.states.mean, sol.states.std

fig, (ax1, ax2) = plt.subplots(nrows=2, ncols=1, sharex=True)
ax1.plot(sol.locations, means[:, 0], label="x1")
ax1.plot(sol.locations, means[:, 1], label="x2")
ax1.plot(sol.locations, means[:, 2], label="x3")
ax1.legend()
ax2.fill_between(sol.locations, stds[:, 0], alpha=0.25, label="x1-unc.")
ax2.fill_between(sol.locations, stds[:, 1], alpha=0.25, label="x2-unc.")
ax2.fill_between(sol.locations, stds[:, 2], alpha=0.25, label="x3-unc.")
ax2.legend()
fig.suptitle("EKO Solution")
plt.show()
```

EK0 Solution



```
[56]: import numpy as np

from probnum.diffeq import probsolve_ivp

rng = np.random.default_rng(seed=123)

num_samples = 150
locations = np.arange(0.0, 22.1, 0.05)
locations = sol.locations
samples = sol.sample(rng=rng, t=locations, size=num_samples)

solution = sol(locations)
means = solution.mean
stds = solution.std

fig, (ax1) = plt.subplots(nrows=1, ncols=1, sharex=True)
for sample in samples[:5]:
    ax1.plot(locations, sample[:, 0], ":", color="C0", linewidth=0.5, alpha=1)
    ax1.plot(locations, sample[:, 1], ":", color="C1", linewidth=0.5, alpha=1)
    ax1.plot(locations, sample[:, 2], ":", color="C2", linewidth=0.5, alpha=1)
ax1.plot(locations, means[:, 0], label="x1", color="k")
ax1.plot(locations, means[:, 1], label="x2", color="k")
```

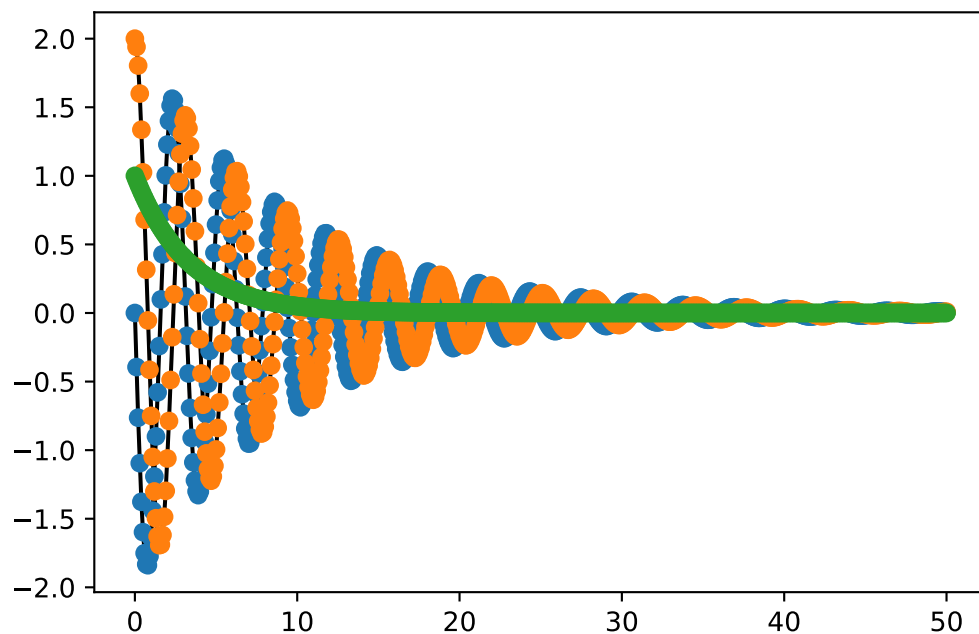
```

ax1.plot(locations, means[:, 2], label="x2", color="k")
ax1.fill_between(
    locations,
    means[:, 0] - 3 * stds[:, 0],
    means[:, 0] + 3 * stds[:, 0],
    color="C0",
    alpha=0.3,
)
ax1.fill_between(
    locations,
    means[:, 1] - 3 * stds[:, 1],
    means[:, 1] + 3 * stds[:, 1],
    color="C1",
    alpha=0.3,
)
ax1.fill_between(
    locations,
    means[:, 2] - 3 * stds[:, 2],
    means[:, 2] + 3 * stds[:, 2],
    color="C2",
    alpha=0.3,
)
ax1.plot(sol.locations, sol.states.mean, "o")
fig.suptitle(f"Samples From an EK0 Solution")

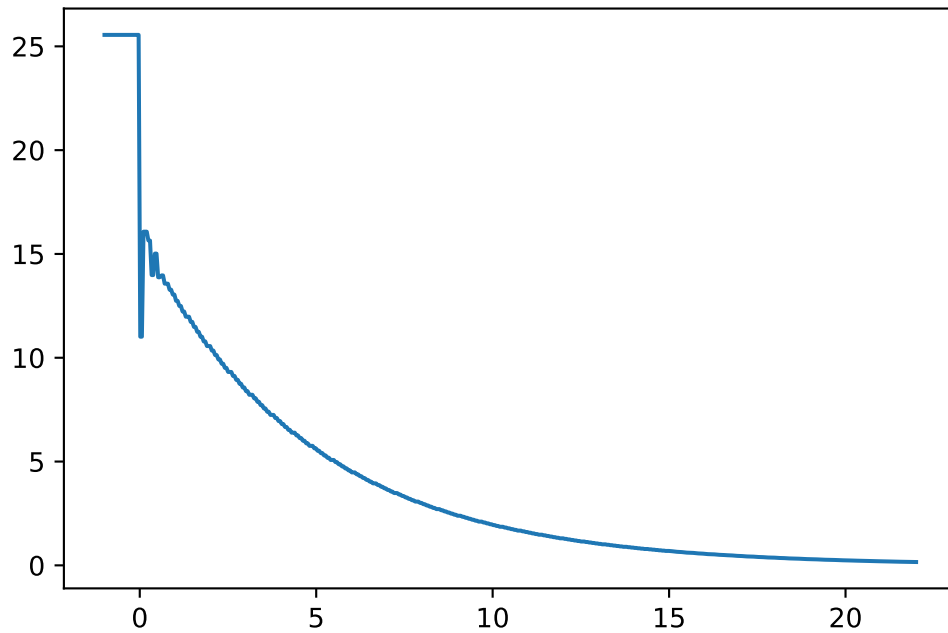
plt.show()

```

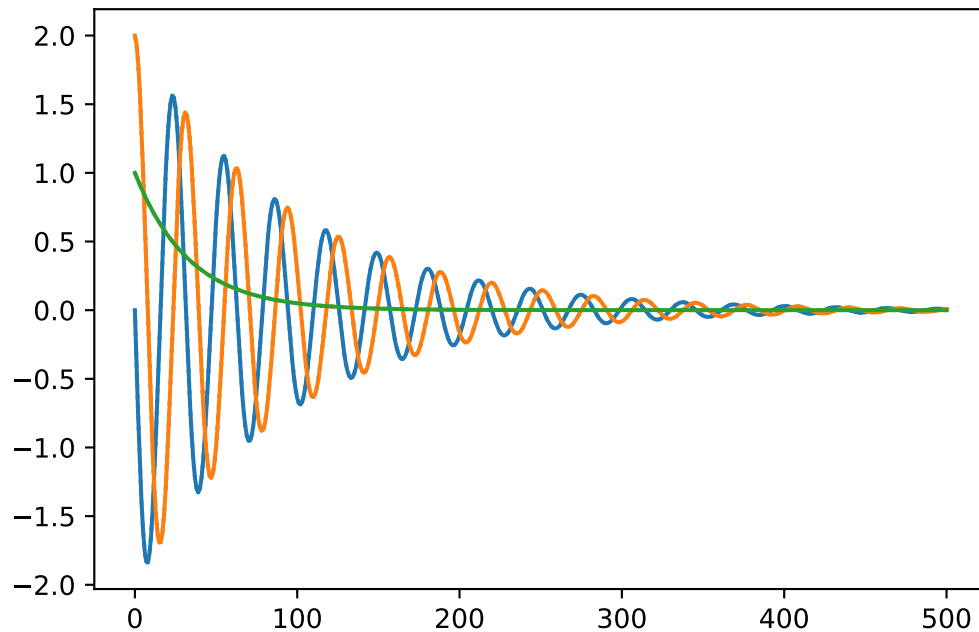
Samples From an EK0 Solution



```
[57]: x = np.linspace(-1, 22, 500)
plt.plot(x, sol.kalman_posterior.diffusion_model(x))
plt.show()
```



```
[58]: plt.plot(samples.mean(axis=0), color="gray", linestyle="dotted")
plt.plot(means)
plt.show()
```

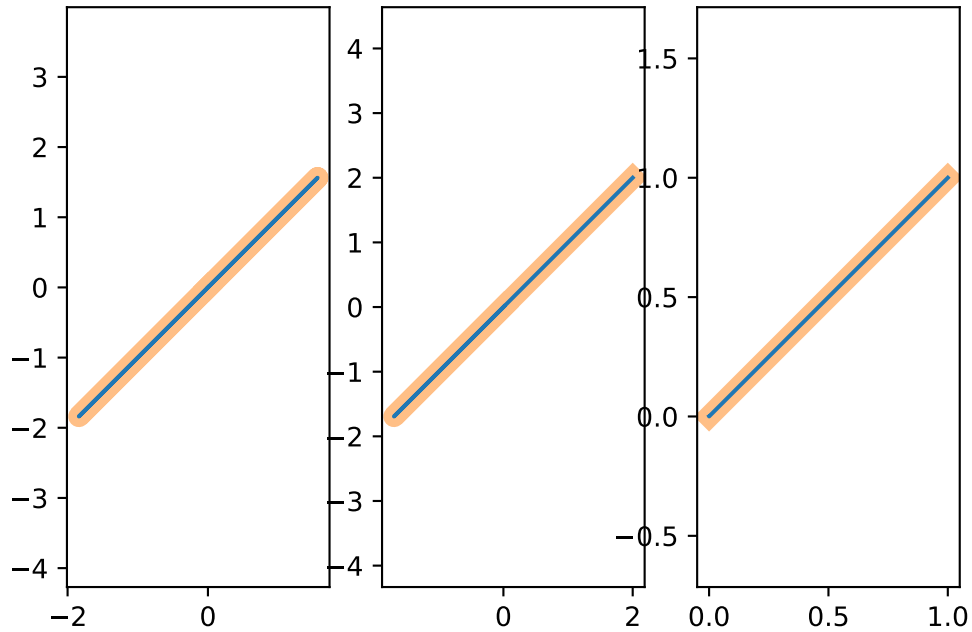


```
[59]: sample_mean = samples.mean(axis=0)
      sample_std = samples.std(axis=0)

      fig, (ax1, ax2, ax3) = plt.subplots(ncols=3)

      ax1.plot(means[:, 0], sample_mean[:, 0])
      ax2.plot(means[:, 1], sample_mean[:, 1])
      ax3.plot(means[:, 2], sample_mean[:, 2])
      ax1.plot(means[:, 0], means[:, 0], alpha=0.5, linewidth=8, zorder=0)
      ax2.plot(means[:, 1], means[:, 1], alpha=0.5, linewidth=8, zorder=0)
      ax3.plot(means[:, 2], means[:, 2], alpha=0.5, linewidth=8, zorder=0)

      ax1.axis("equal")
      ax2.axis("equal")
      ax3.axis("equal")
      plt.show()
```

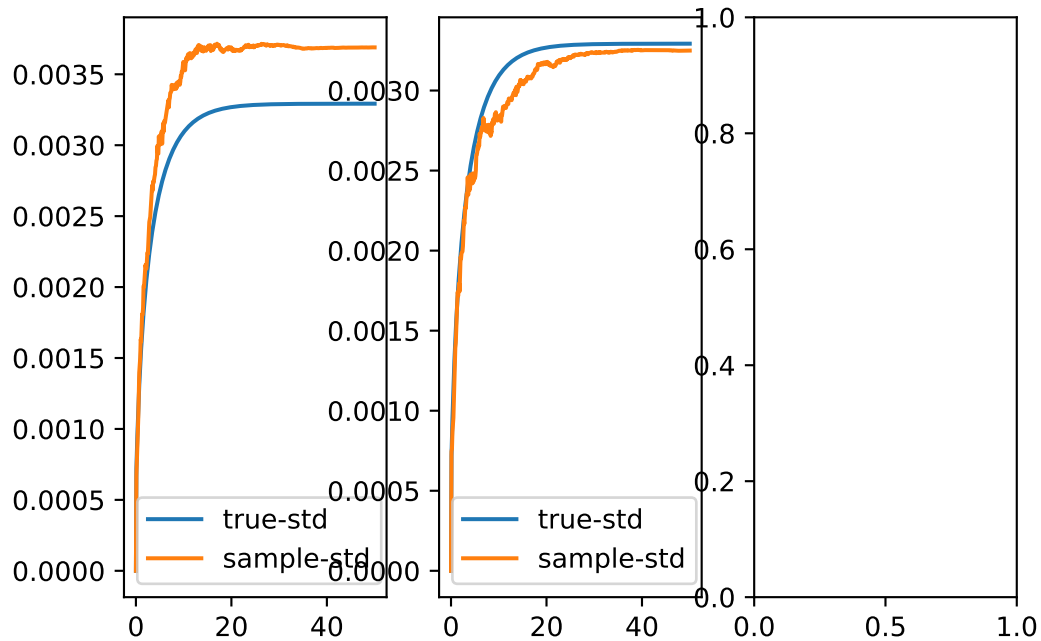


```
[60]: fig, (ax1, ax2, ax3) = plt.subplots(ncols=3)

ax1.plot(locations, stds[:, 0], label="true-std")
ax1.plot(locations, sample_std[:, 0], label="sample-std")
ax1.legend()

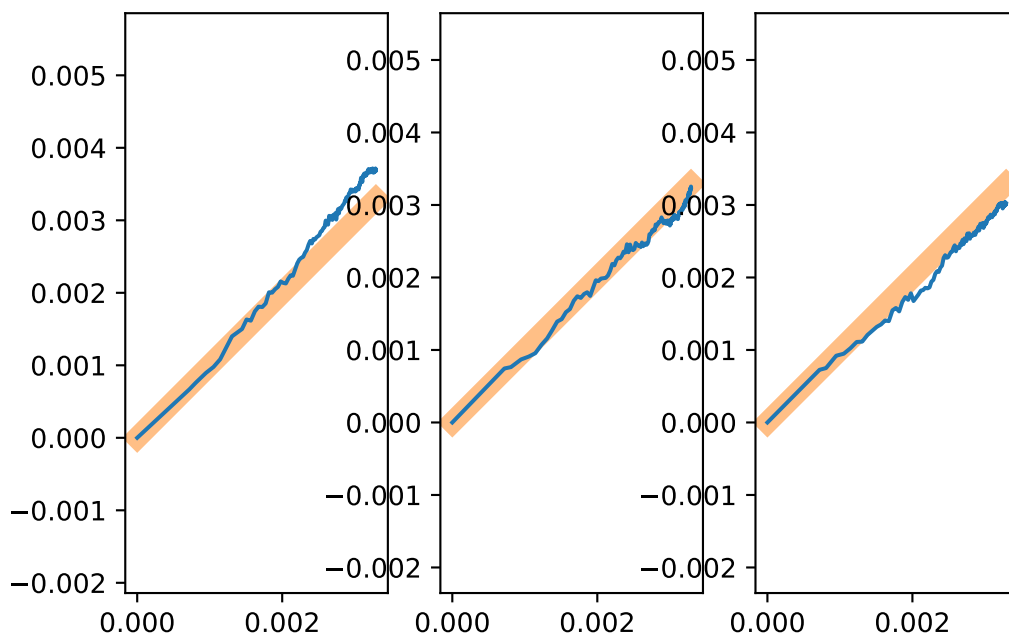
ax2.plot(locations, stds[:, 1], label="true-std")
ax2.plot(locations, sample_std[:, 1], label="sample-std")
ax2.legend()
plt.show()

ax3.plot(locations, stds[:, 2], label="true-std")
ax3.plot(locations, sample_std[:, 2], label="sample-std")
ax3.legend()
plt.show()
```



```
[61]: fig, (ax1, ax2, ax3) = plt.subplots(ncols=3)

ax1.plot(stds[:, 0], sample_std[:, 0])
ax2.plot(stds[:, 1], sample_std[:, 1])
ax3.plot(stds[:, 2], sample_std[:, 2])
ax1.plot(stds[:, 0], stds[:, 0], alpha=0.5, linewidth=8, zorder=0)
ax2.plot(stds[:, 1], stds[:, 1], alpha=0.5, linewidth=8, zorder=0)
ax3.plot(stds[:, 2], stds[:, 2], alpha=0.5, linewidth=8, zorder=0)
ax1.axis("equal")
ax2.axis("equal")
ax3.axis("equal")
plt.show()
```



[]: